NYU

Missing Persons

Ayush Sarkar
New York University: UGSRP
Center for Urban Science + Progress (CUSP)

Cumulative NamUs Cases

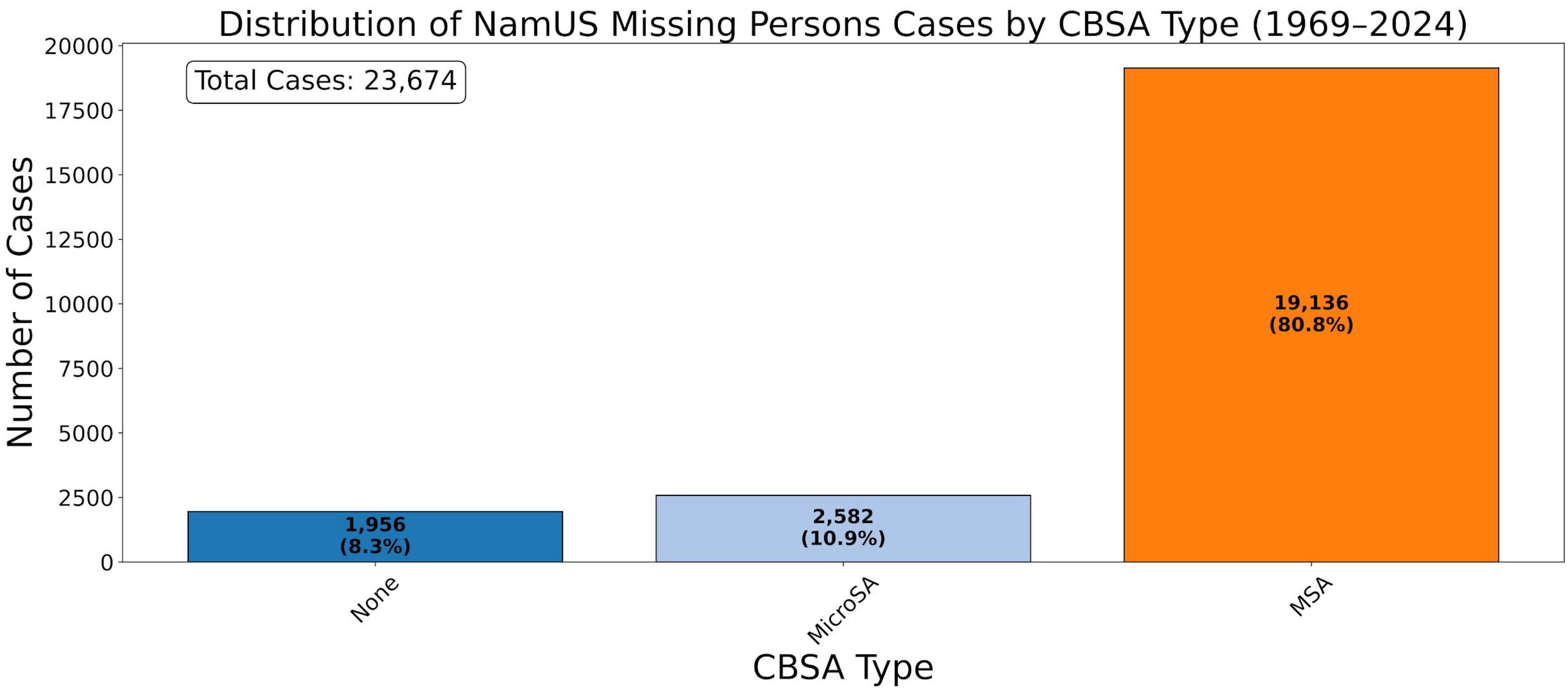
- Scaling of Cumulative Cases vs Population
- Choropleths
- Demographic Analysis of Cases
    - Population Pyramids
    - Sex Distribution
    - Race/Ethnicity Distribution

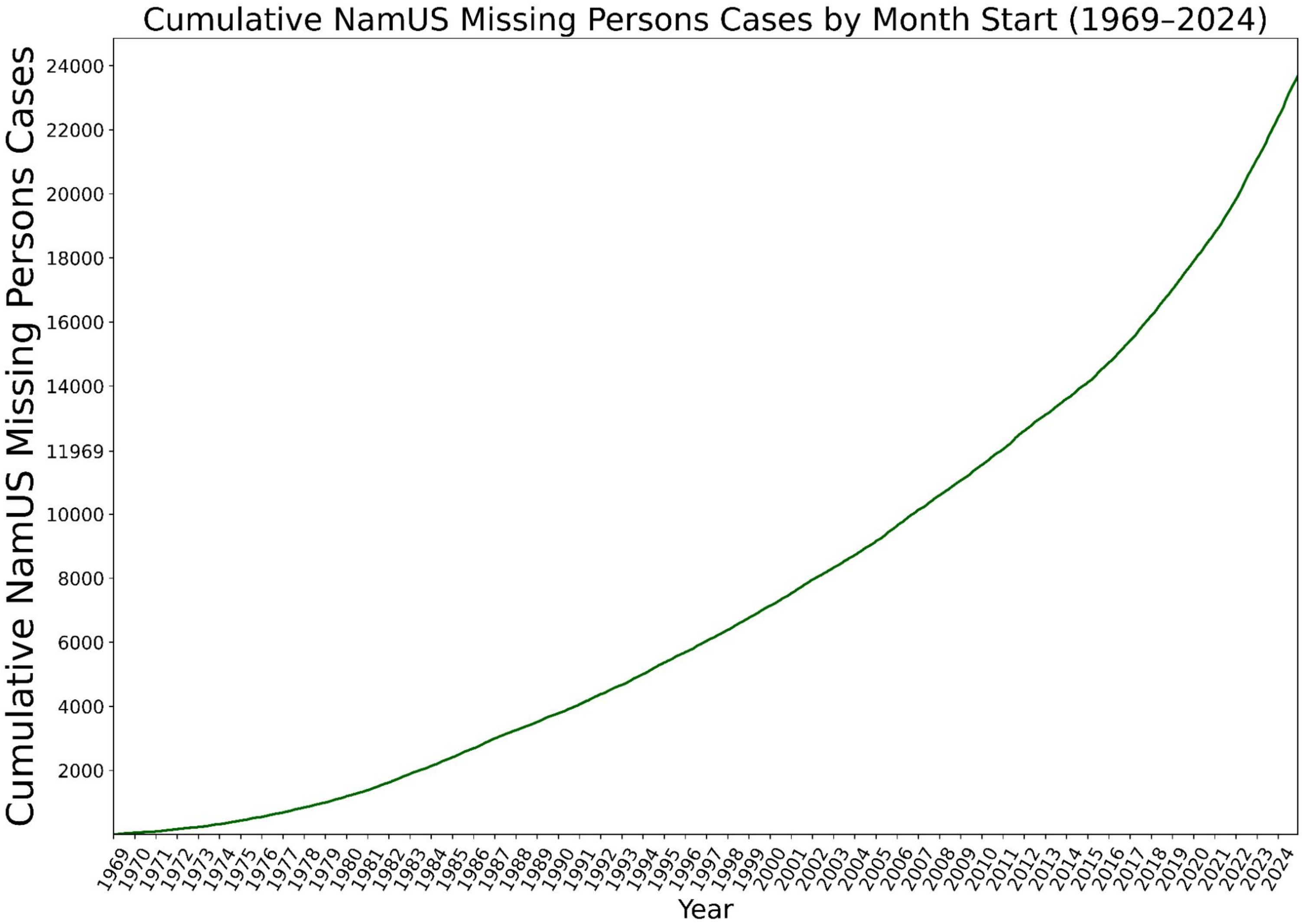
NYU

NamUs Missing Persons Cases Data Characteristics

- We have **25,630** available records and use **16,370** records (from 2000-2024), within the decades:
  - 1900-1910: **1**
  - 1910-1920: **2**
  - 1920-1930: **6**
  - 1930-1940: **8**
  - 1940-1950: **30**
  - 1950-1960: **86**
  - 1960-1970: **268**
  - 1970-1980: **1272**
  - 1980-1990: **2961**
  - 1990-2000: **3534**
  - 2000-2010: **4465**
  - 2010-2020: **6396**
  - 2020-2024: **5899**
  - 2025: **702**

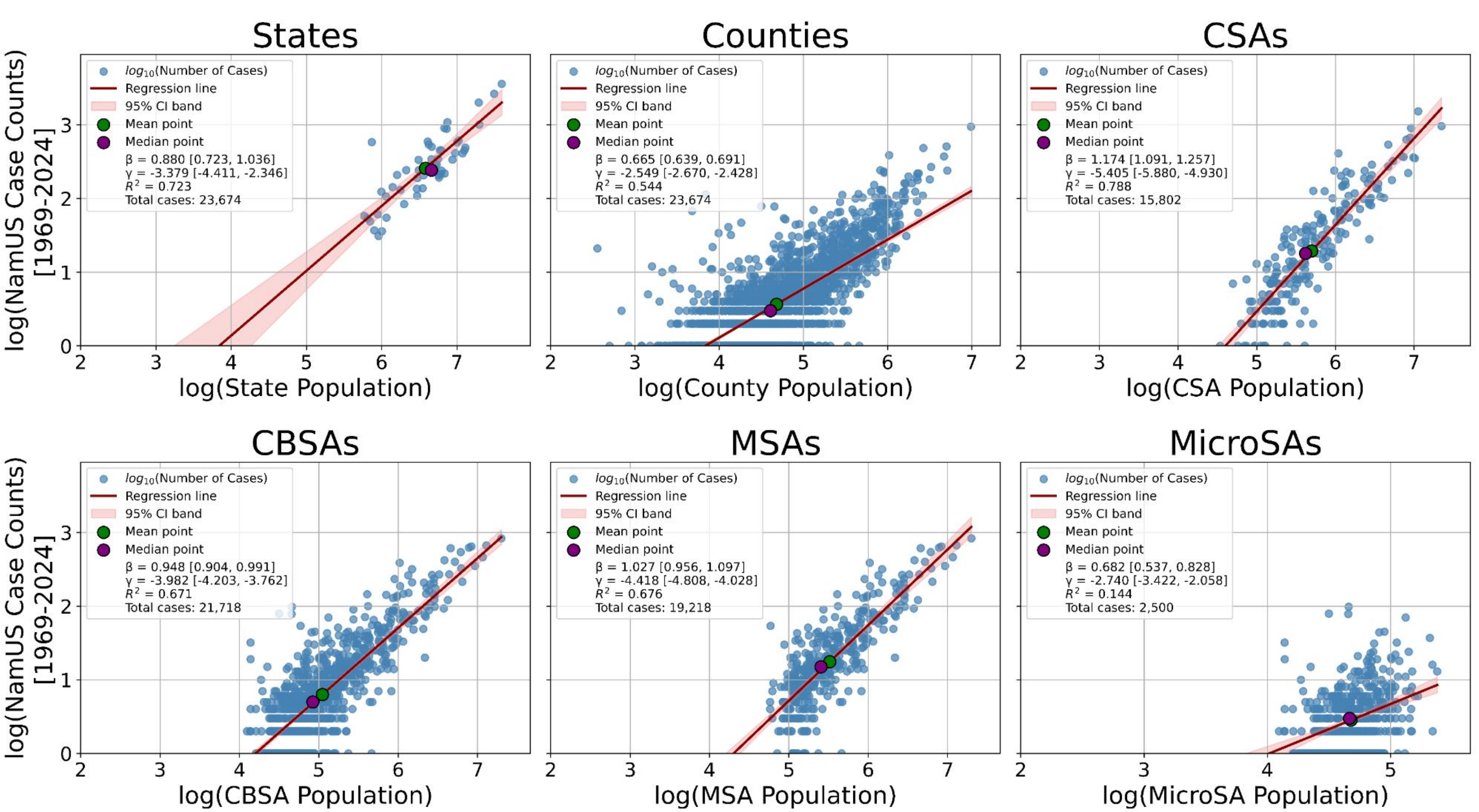
- The data missing per column is as follows:
  - CaseID: **0.00%**
  - FirstName: **0.00%**
  - MiddleName: **24.78%**
  - LastName: **0.00%**
  - Nicknames: **72.19%**
  - CurrentMinAge: **0.00%**
  - CurrentMaxAge: **0.00%**
  - Sex: **0.00%**
  - Ethnicity: **0.00%**
  - HeightInches: **0.14%**
  - WeightPounds: **0.19%**
  - EyeColor: **2.41%**
  - HairColor: **1.76%**
  - DisappearanceDate: **0.00%**
  - City: **0.42%**
  - State: **0.00%**
  - County: **0.31%**
  - InvestigatingAgency: **0.01%**


Distribution of NamUS Missing Persons Cases by CBSA Type (1969–2024)
Total Cases: 23,674
Number of Cases
20000
17500
15000
12500
10000
7500
5000
2500
0
1,956
(8.3%)
2,582
(10.9%)
19,136
(80.8%)
None
MicroSA
MSA
CBSA Type

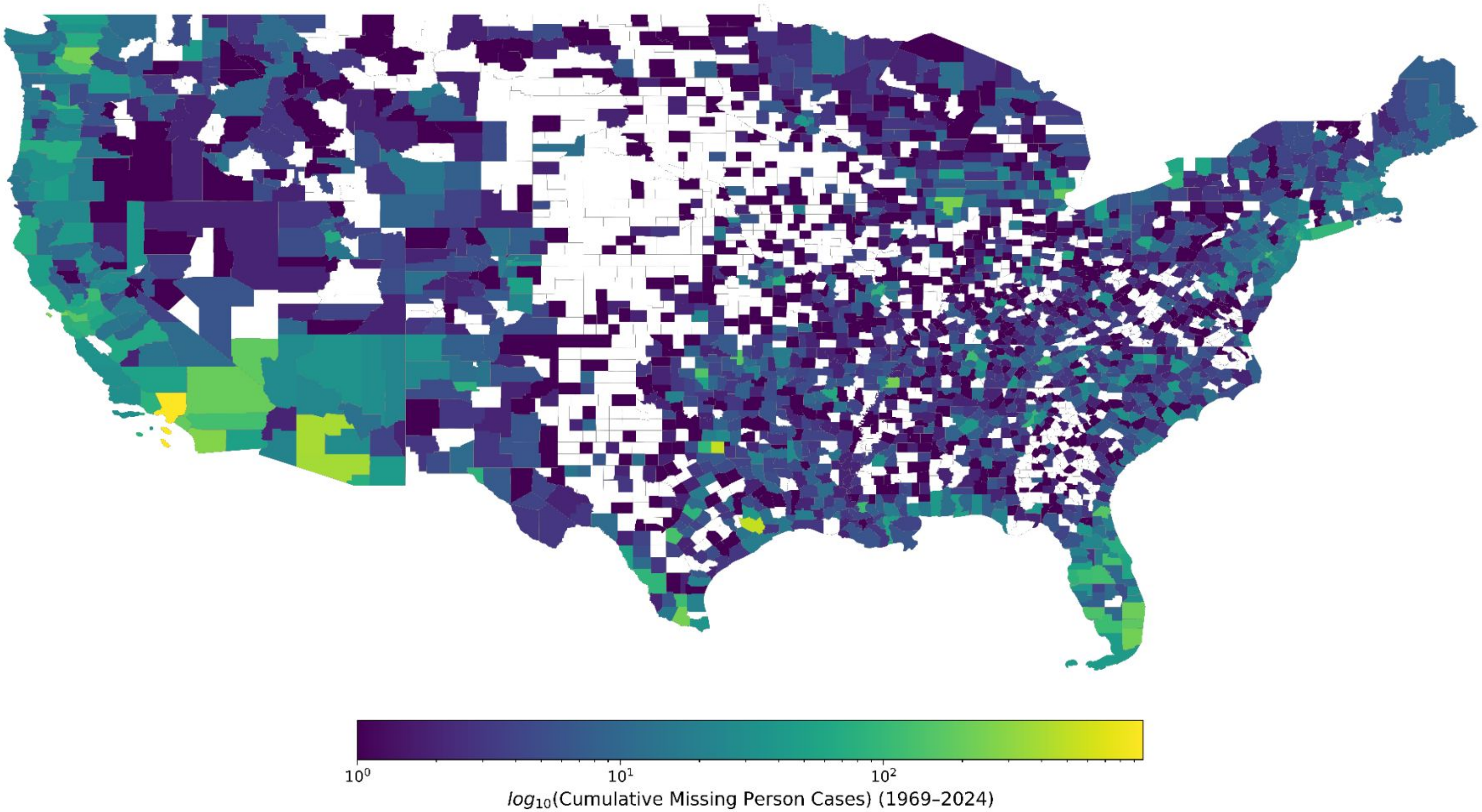

Cumulative NamUS Missing Persons Cases by Month Start (1969–2024)
Cumulative NamUS Missing Persons Cases
24000
22000
20000
18000
16000
14000
11969
10000
8000
6000
4000
2000
1969 1970 1971 1972 1973 1974 1975 1976 1977 1978 1979 1980 1981 1982 1983 1984 1985 1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024
Year

NY

Scaling Exponent (β) of NamUs Missing Persons Cases vs GEOID Population [1969–2024]

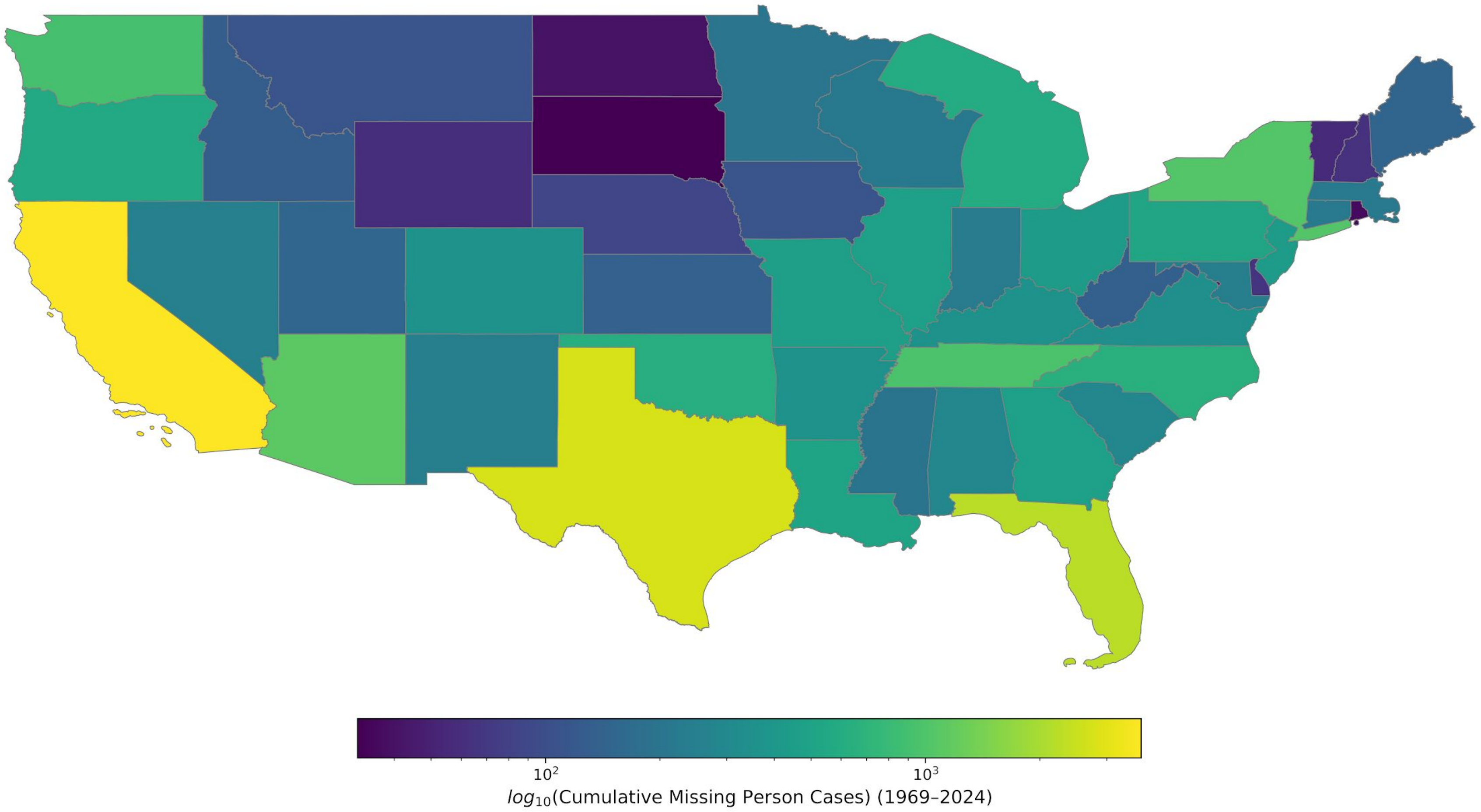


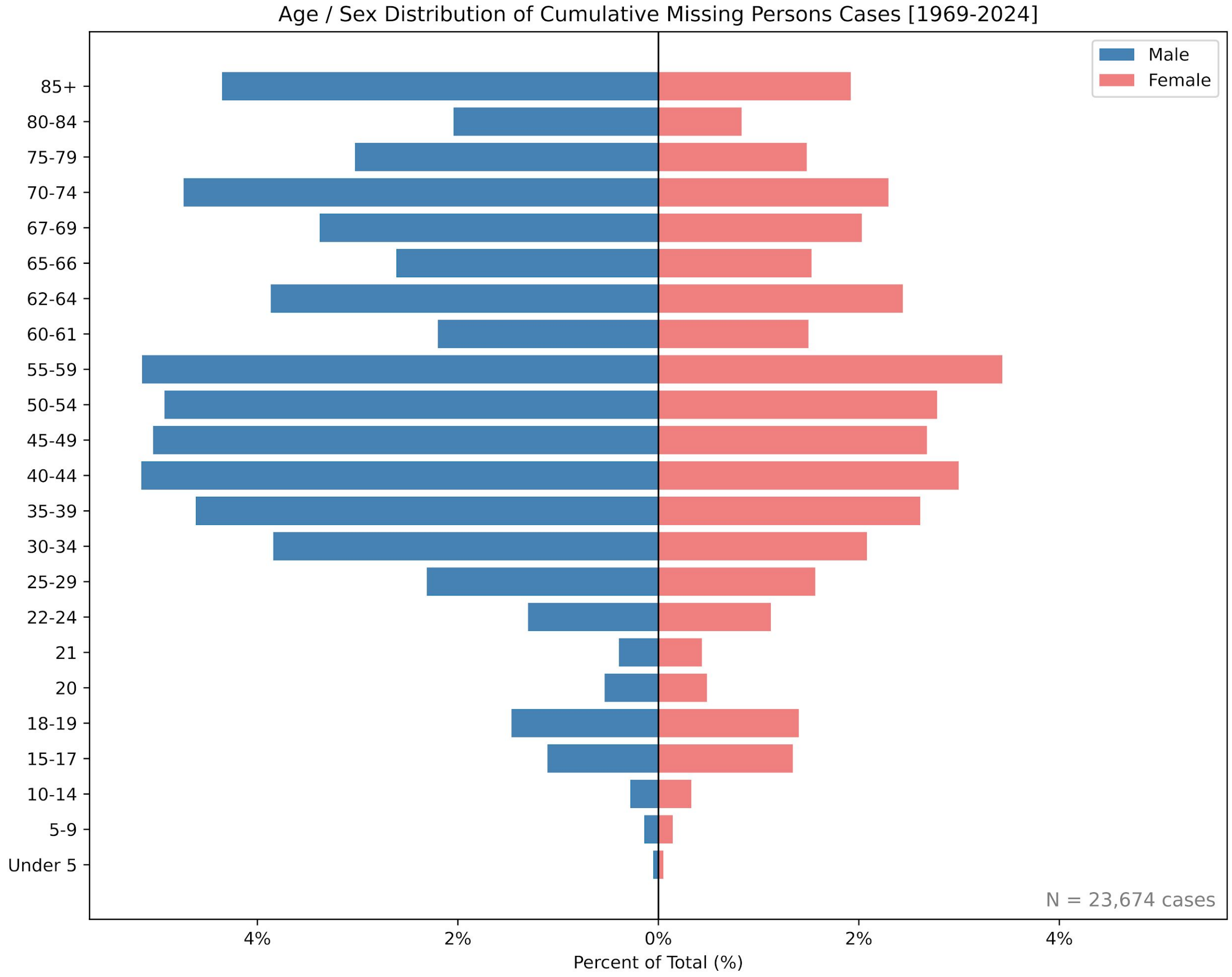
Cumulative NamUs Missing Person Cases by County (Continental U.S., 1969–2024)



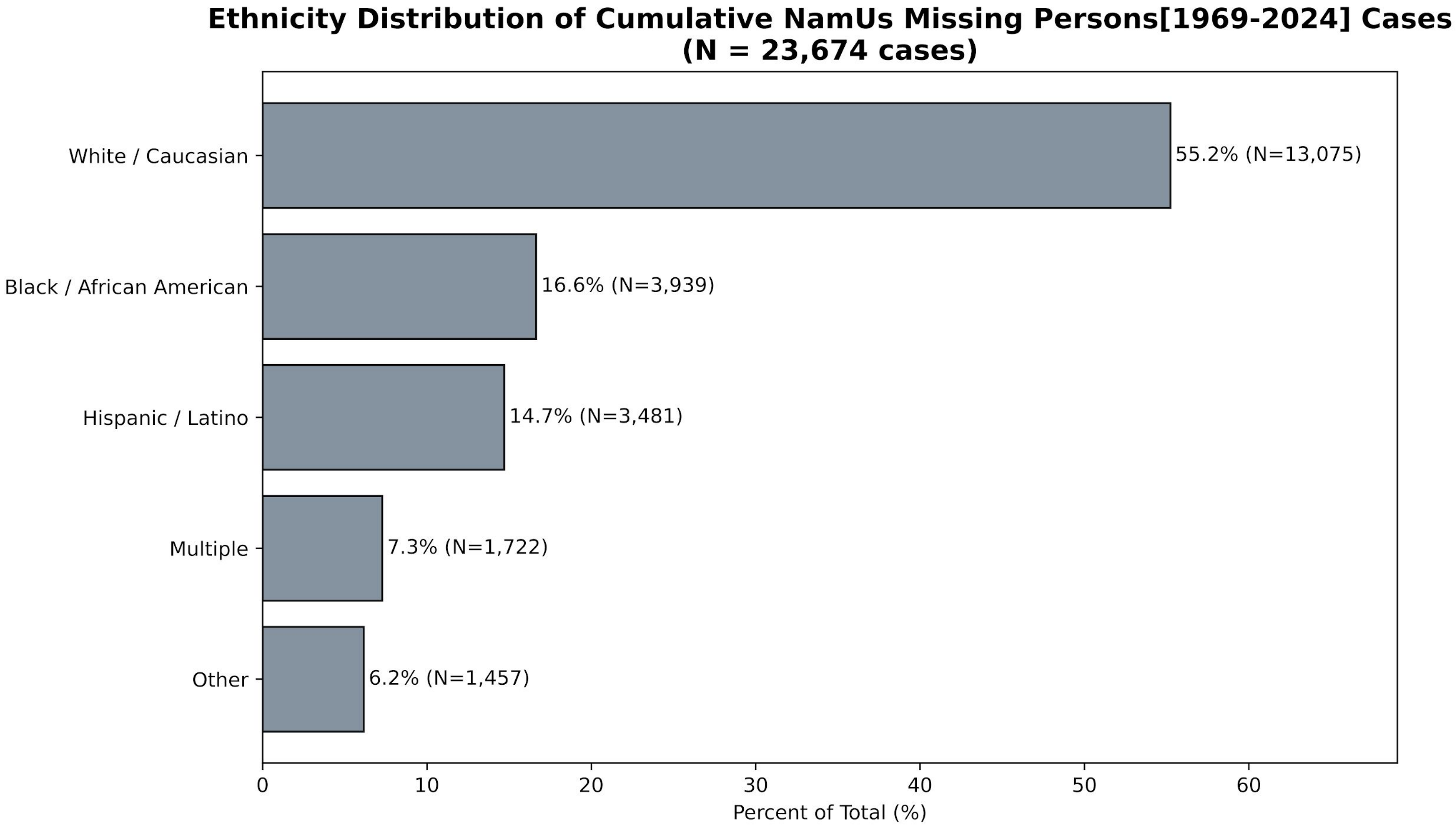
log_{10}(Cumulative Missing Person Cases) (1969–2024)

Cumulative NamUs Missing Person Cases by State (Continental U.S., 1969–2024)

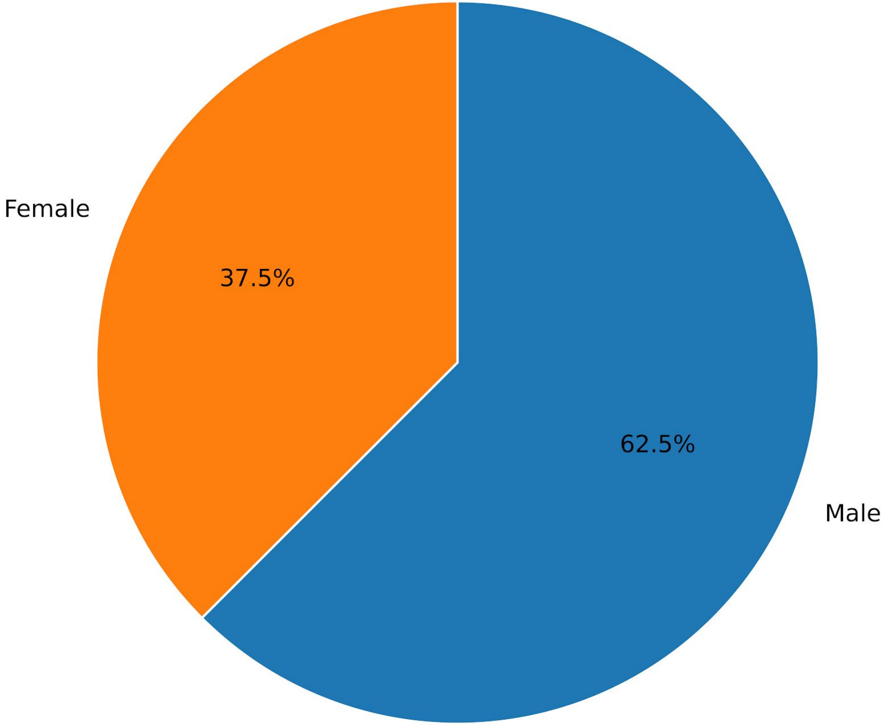



Age / Sex Distribution of Cumulative Missing Persons Cases [1969-2024]
Male
Female
85+
80-84
75-79
70-74
67-69
65-66
62-64
60-61
55-59
50-54
45-49
40-44
35-39
30-34
25-29
22-24
21
20
18-19
15-17
10-14
5-9
Under 5
N = 23,674 cases
4%
2%
0%
2%
4%
Percent of Total (%)

NYU


Ethnicity Distribution of Cumulative NamUs Missing Persons[1969-2024] Cases
(N = 23,674 cases)
White / Caucasian
55.2% (N=13,075)
Black / African American
16.6% (N=3,939)
Hispanic / Latino
14.7% (N=3,481)
Multiple
7.3% (N=1,722)
Other
6.2% (N=1,457)
0
10
20
30
40
50
60
Percent of Total (%)

Sex Distribution of Cumulative NamUs Missing Persons[1969-2024] Cases
(N = 23,674 cases)



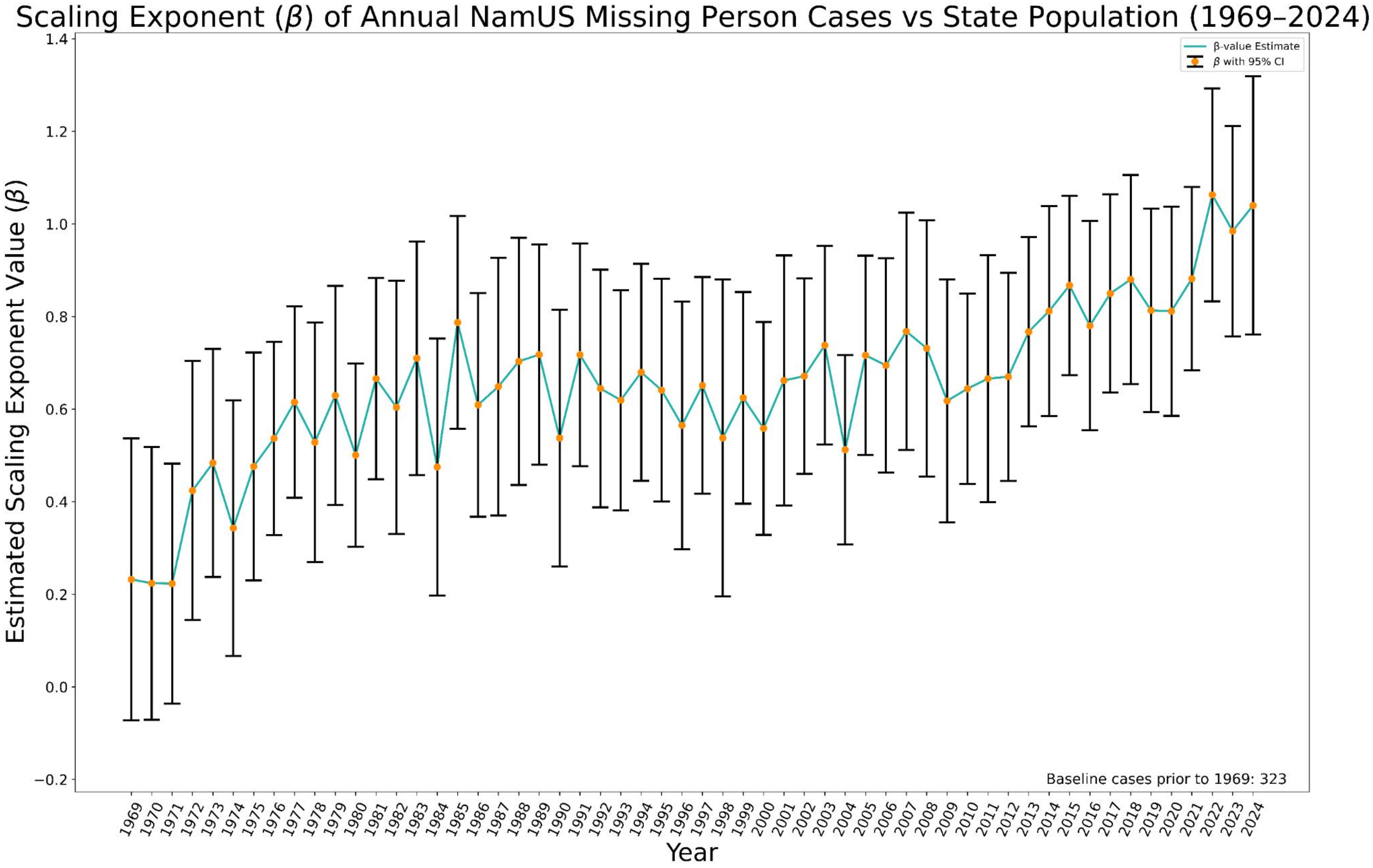
Annual NamUs Cases

- Time-series of Scaling Exponent of Annual Cases vs GEOID Population from [1969-2024]
    - Counties
    - CSAs
    - CBSAs; MSAs and MicroSAs
- R-squared Time Series [1969-2024]
- Best and Worst Year R-squared Regression Plots

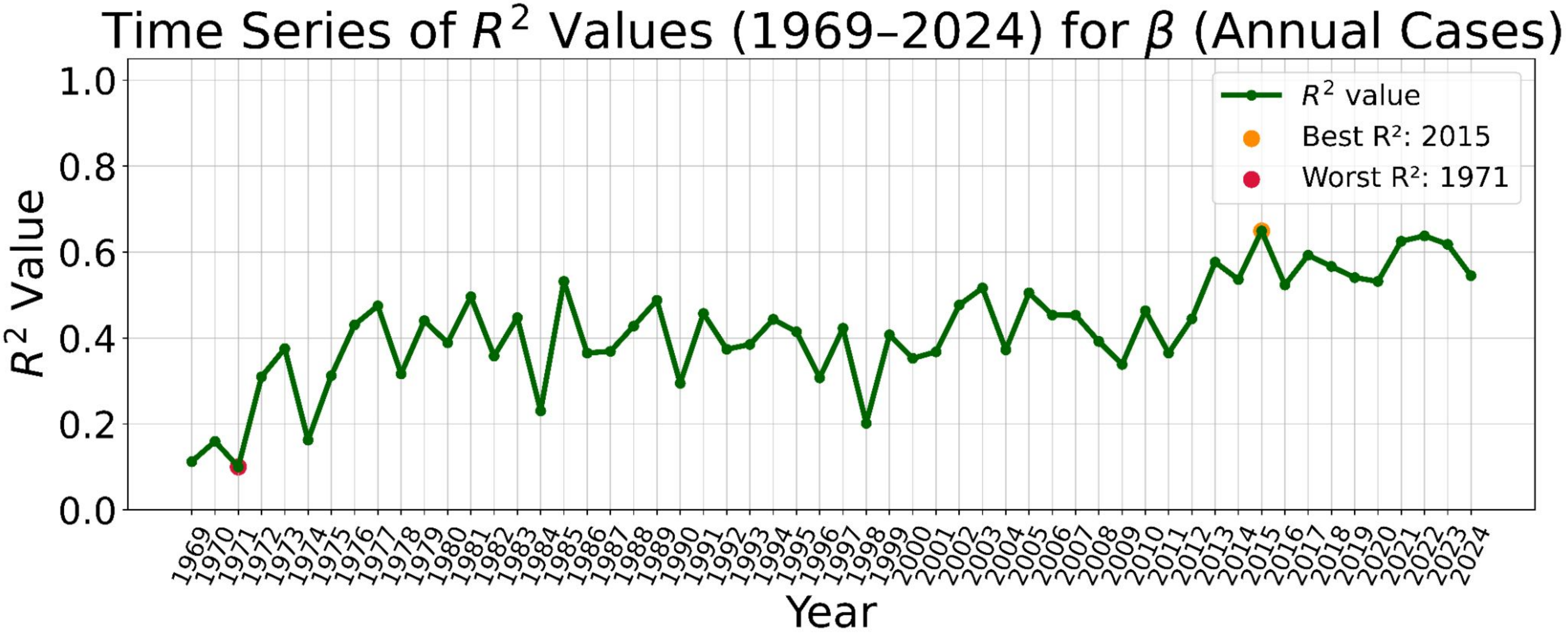
NYU

State Level

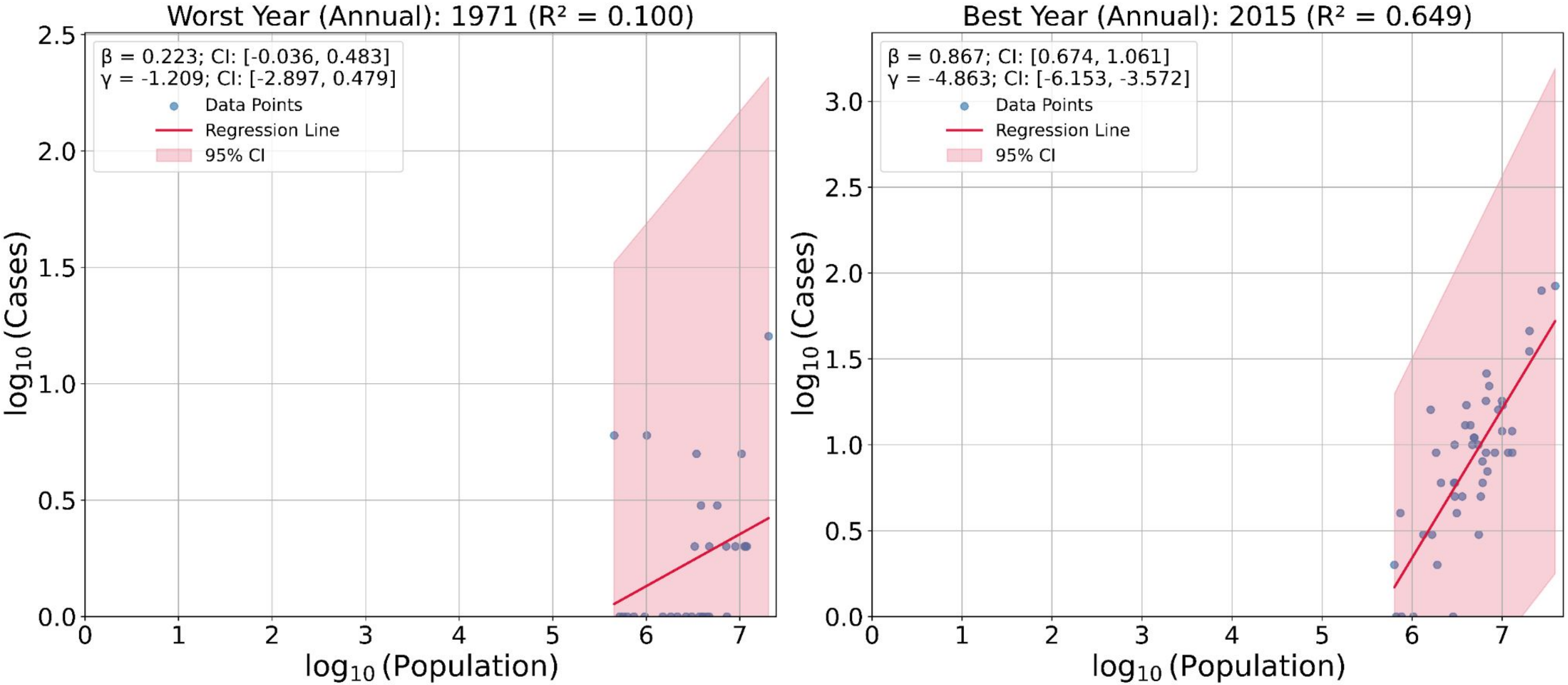
NYU



Scaling Exponent (β) of Annual NamUS Missing Person Cases vs State Population (1969–2024)
β-value Estimate
β with 95% CI
Estimated Scaling Exponent Value (β)
Year
Baseline cases prior to 1969: 323


Time Series of R^2 Values (1969–2024) for β (Annual Cases)
R^2 value
Best R²: 2015
Worst R²: 1971
R^2 Value
Year
1.0
0.8
0.6
0.4
0.2
0.0
1969
1970
1971
1972
1973
1974
1975
1976
1977
1978
1979
1980
1981
1982
1983
1984
1985
1986
1987
1988
1989
1990
1991
1992
1993
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024

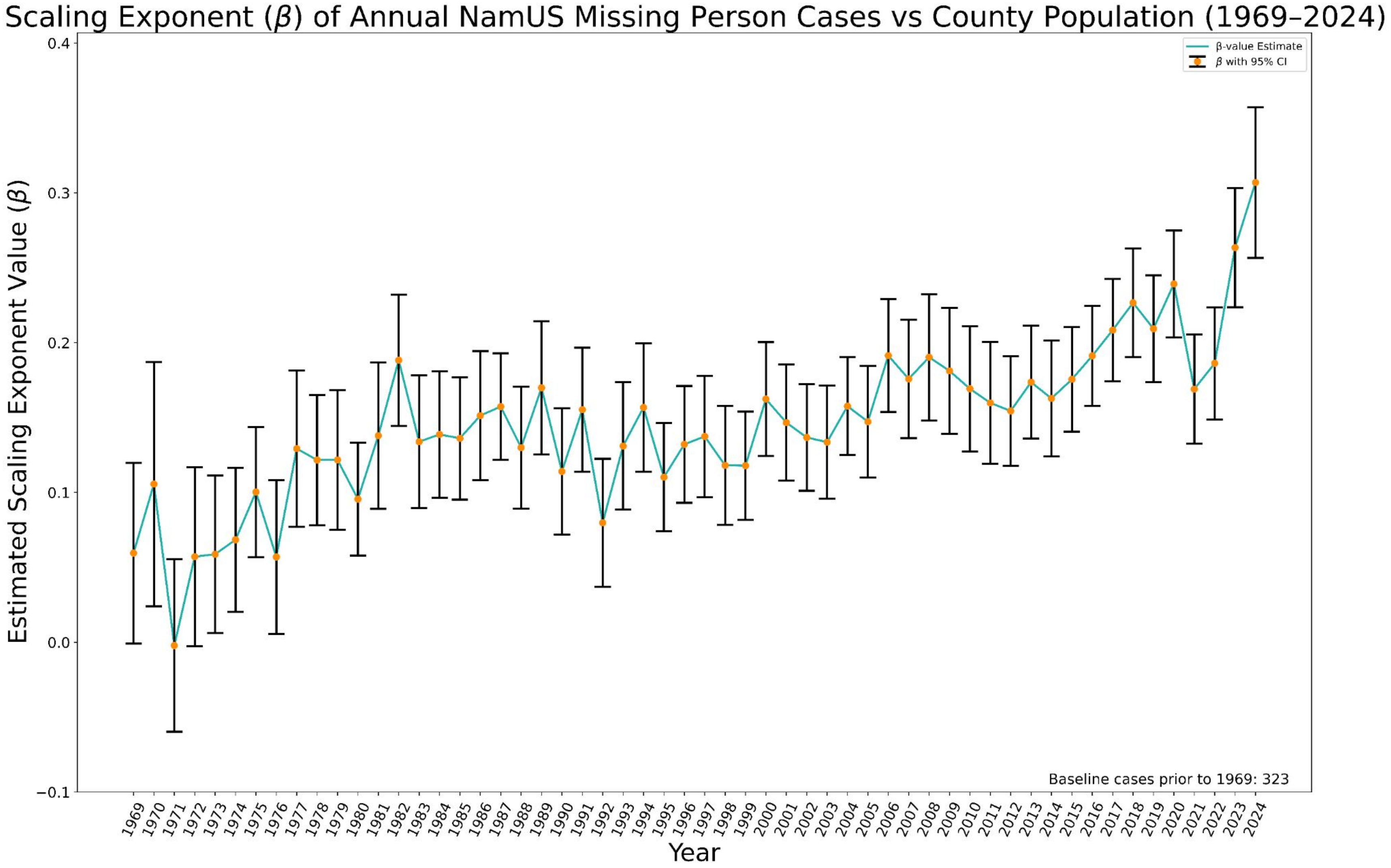
NYU

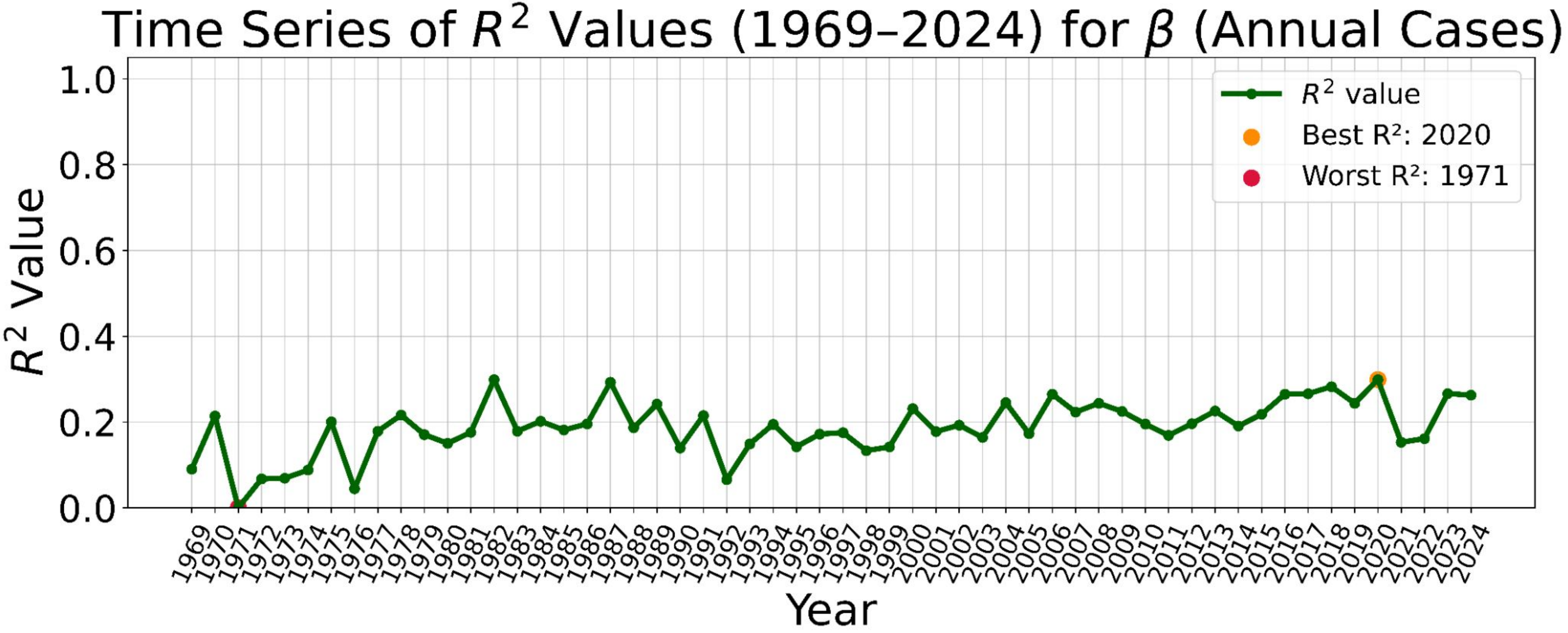

Worst Year (Annual): 1971 (R^2 = 0.100)
β = 0.223; CI: [-0.036, 0.483]
γ = -1.209; CI: [-2.897, 0.479]
Data Points
Regression Line
95% CI
$\log_{10}$(Cases)
$\log_{10}$(Population)
Best Year (Annual): 2015 (R^2 = 0.649)
β = 0.867; CI: [0.674, 1.061]
γ = -4.863; CI: [-6.153, -3.572]
Data Points
Regression Line
95% CI
$\log_{10}$(Cases)
$\log_{10}$(Population)

NYU

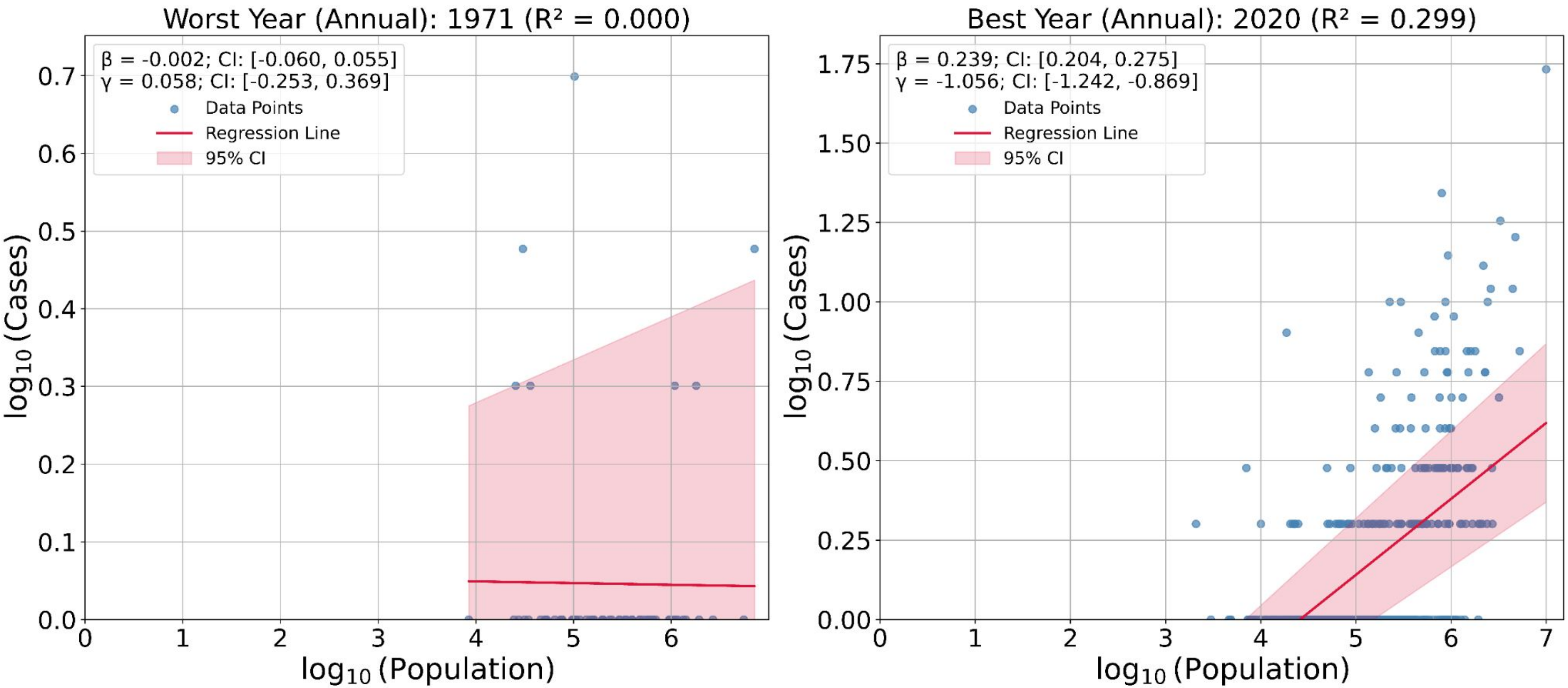
County-level

NYU


Scaling Exponent (β) of Annual NamUS Missing Person Cases vs County Population (1969–2024)
β-value Estimate
β with 95% CI
Estimated Scaling Exponent Value (β)
Year
Baseline cases prior to 1969: 323



Time Series of R^2 Values (1969–2024) for β (Annual Cases)
R^2 value
Best R²: 2020
Worst R²: 1971
R^2 Value
1.0
0.8
0.6
0.4
0.2
0.0
1969 1970 1971 1972 1973 1974 1975 1976 1977 1978 1979 1980 1981 1982 1983 1984 1985 1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024
Year

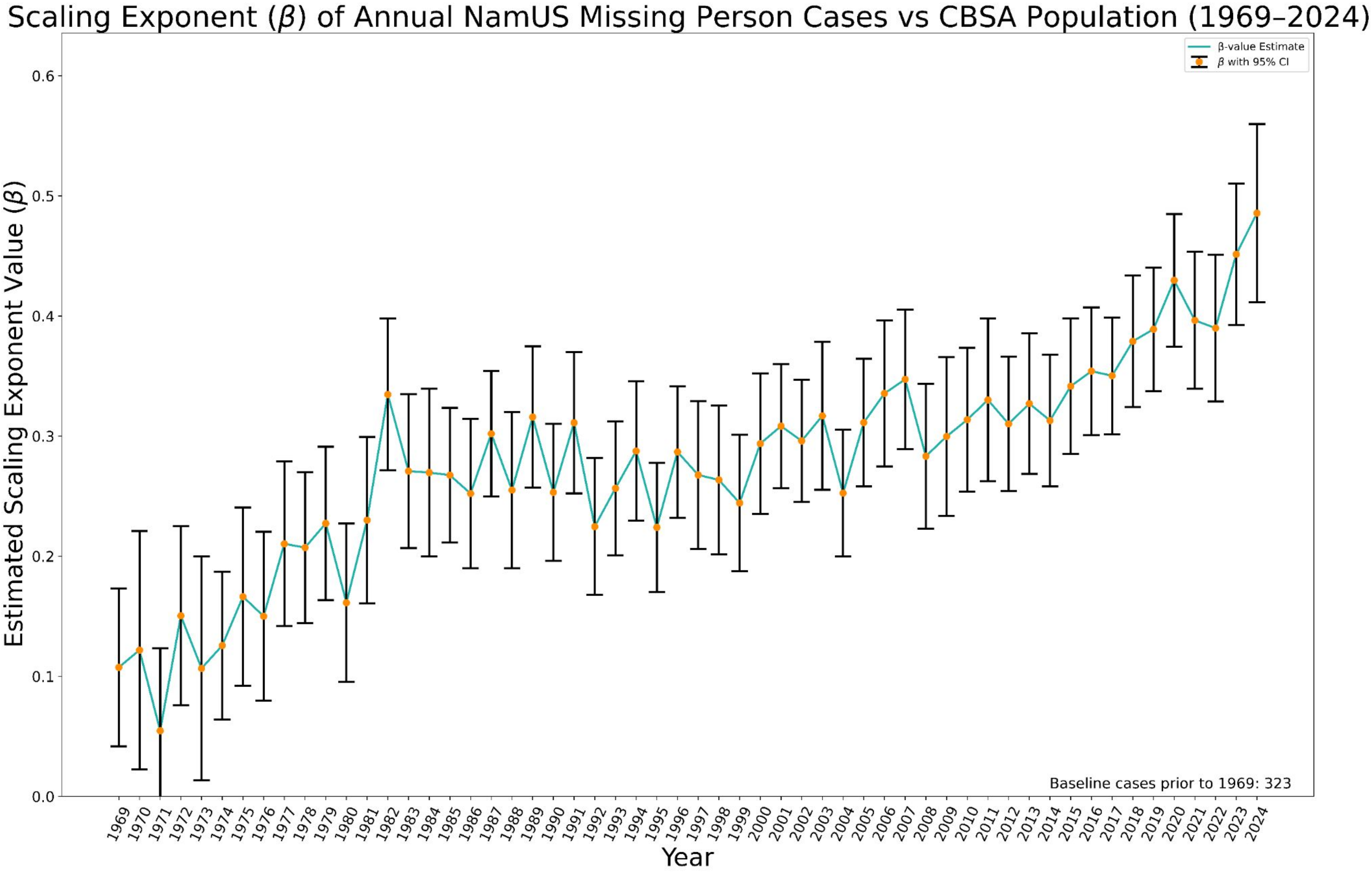


Worst Year (Annual): 1971 (R² = 0.000)
β = -0.002; CI: [-0.060, 0.055]
γ = 0.058; CI: [-0.253, 0.369]
Data Points
Regression Line
95% CI
log10 (Cases)
log10 (Population)
Best Year (Annual): 2020 (R² = 0.299)
β = 0.239; CI: [0.204, 0.275]
γ = -1.056; CI: [-1.242, -0.869]
Data Points
Regression Line
95% CI
log10 (Cases)
log10 (Population)

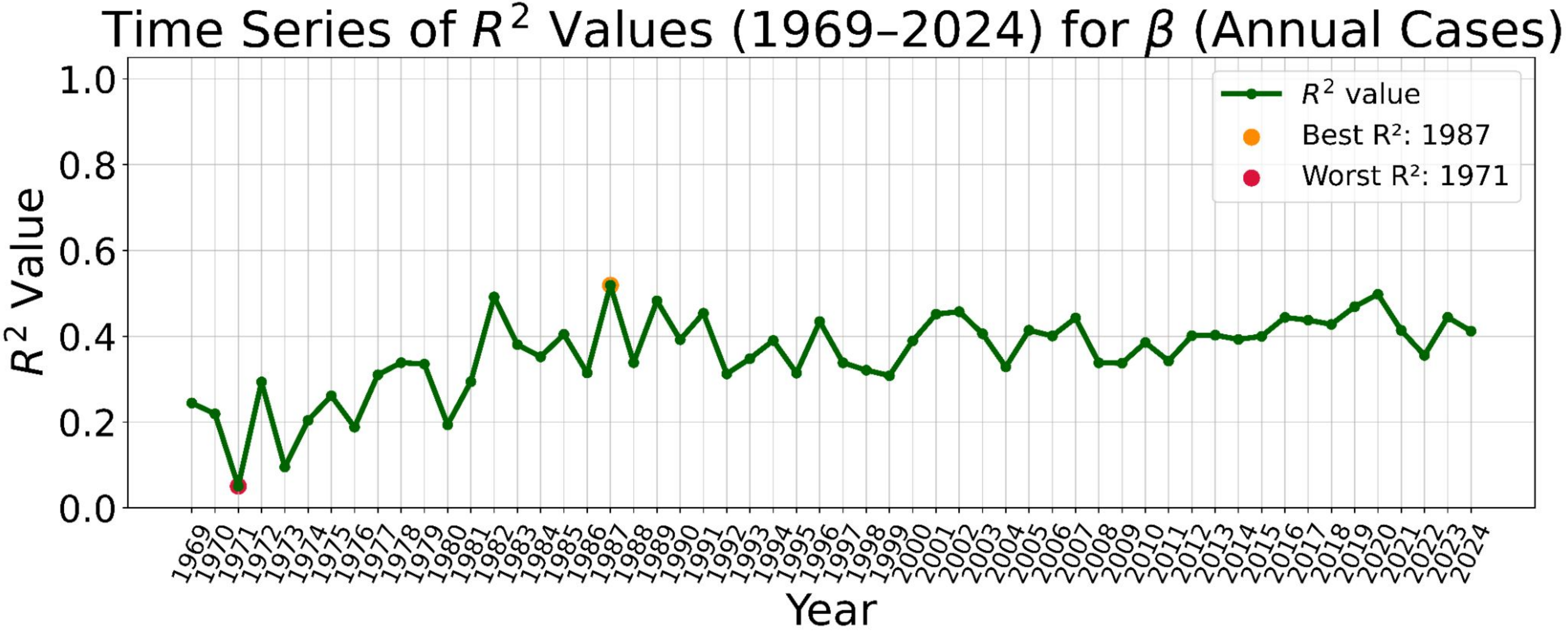
NYU

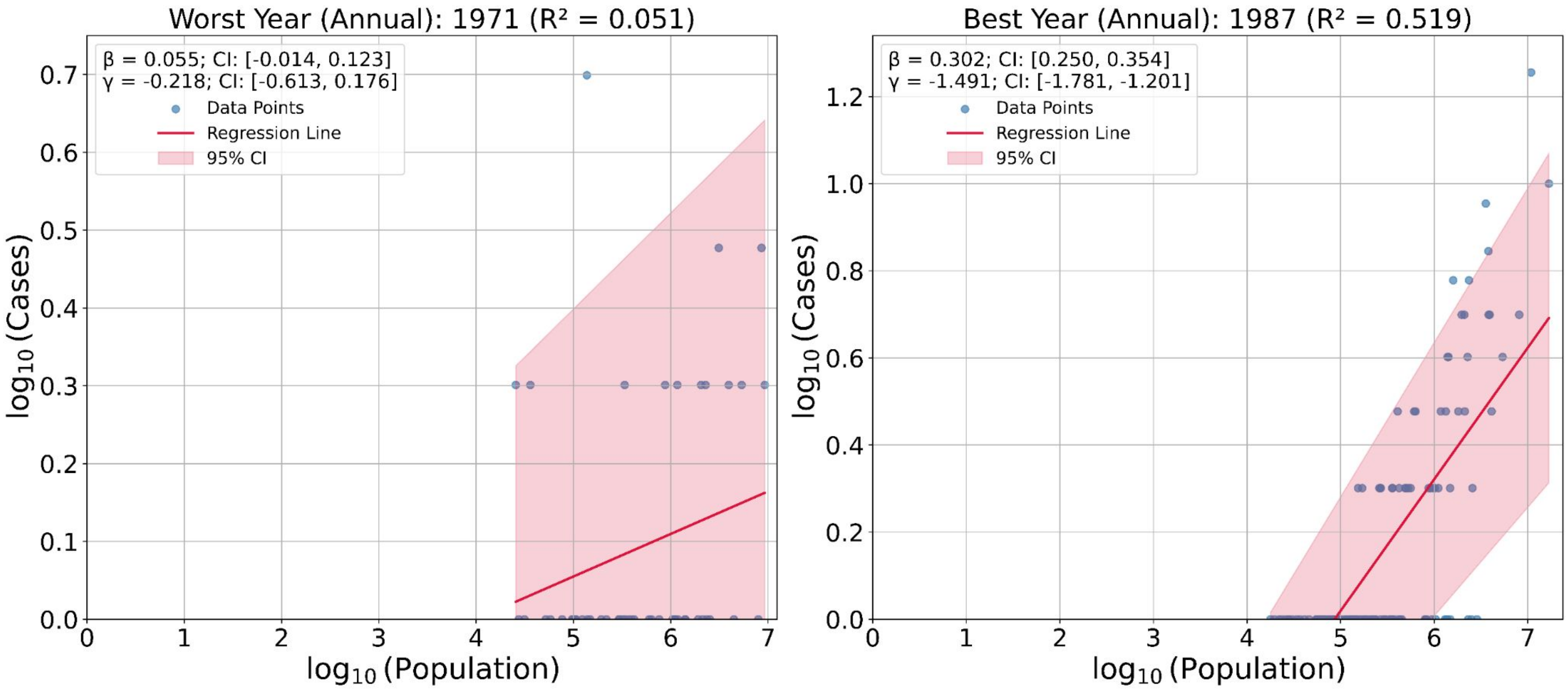
CBSA-level

NYU



Scaling Exponent (β) of Annual NamUS Missing Person Cases vs CBSA Population (1969–2024)
β-value Estimate
β with 95% CI
Estimated Scaling Exponent Value (β)
Year
Baseline cases prior to 1969: 323

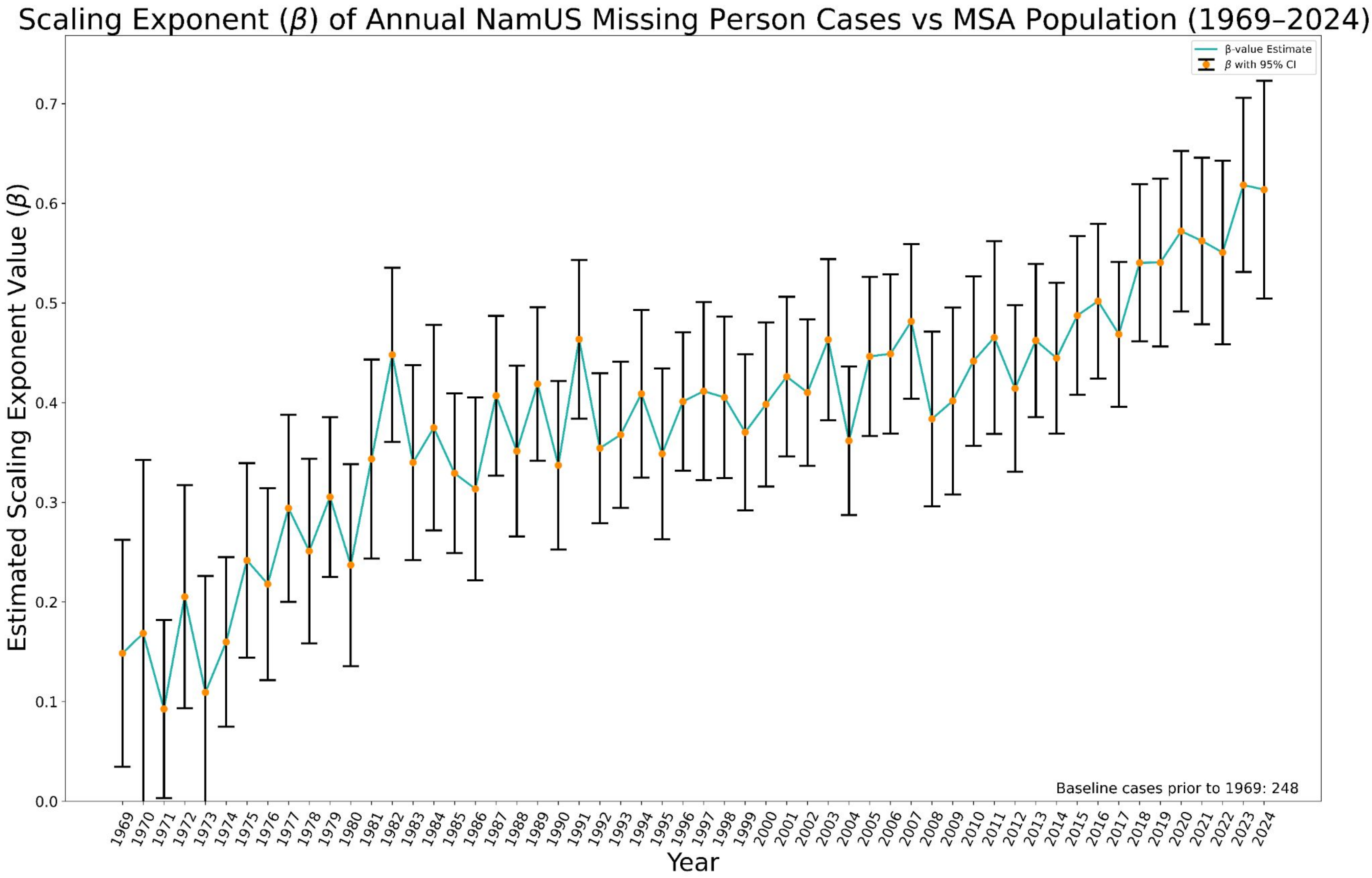

Time Series of R^2 Values (1969–2024) for β (Annual Cases)
R^2 value
Best R²: 1987
Worst R²: 1971
R^2 Value
Year
0.0
0.2
0.4
0.6
0.8
1.0

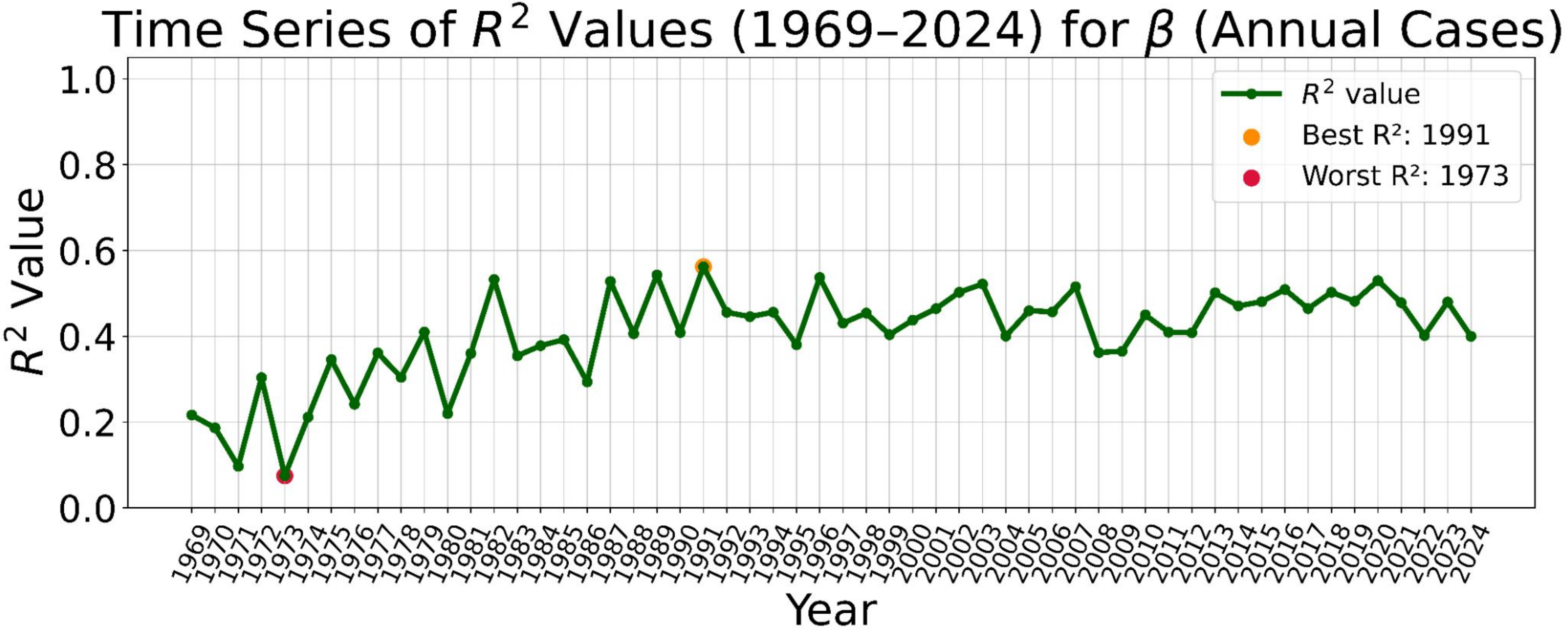

Worst Year (Annual): 1971 (R² = 0.051)
β = 0.055; CI: [-0.014, 0.123]
γ = -0.218; CI: [-0.613, 0.176]
Data Points
Regression Line
95% CI
log10(Cases)
log10(Population)
Best Year (Annual): 1987 (R² = 0.519)
β = 0.302; CI: [0.250, 0.354]
γ = -1.491; CI: [-1.781, -1.201]
Data Points
Regression Line
95% CI
log10(Cases)
log10(Population)

NYU

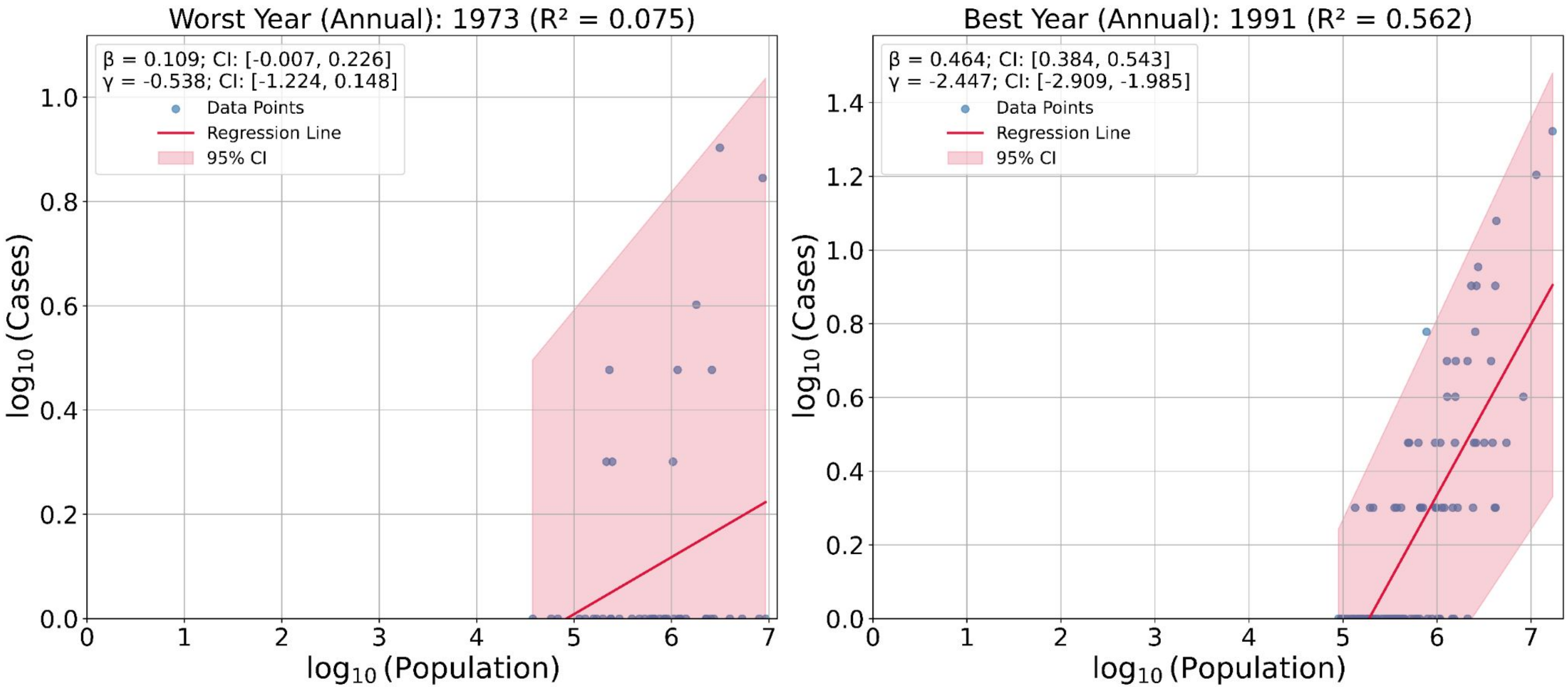
MSA-level

NYU


Scaling Exponent (β) of Annual NamUS Missing Person Cases vs MSA Population (1969–2024)
β-value Estimate
β with 95% CI
Estimated Scaling Exponent Value (β)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Year
1969 1970 1971 1972 1973 1974 1975 1976 1977 1978 1979 1980 1981 1982 1983 1984 1985 1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024
Baseline cases prior to 1969: 248


Time Series of R^2 Values (1969–2024) for β (Annual Cases)
R^2 value
Best R²: 1991
Worst R²: 1973
R^2 Value
Year
1.0
0.8
0.6
0.4
0.2
0.0
1969
1970
1971
1972
1973
1974
1975
1976
1977
1978
1979
1980
1981
1982
1983
1984
1985
1986
1987
1988
1989
1990
1991
1992
1993
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024

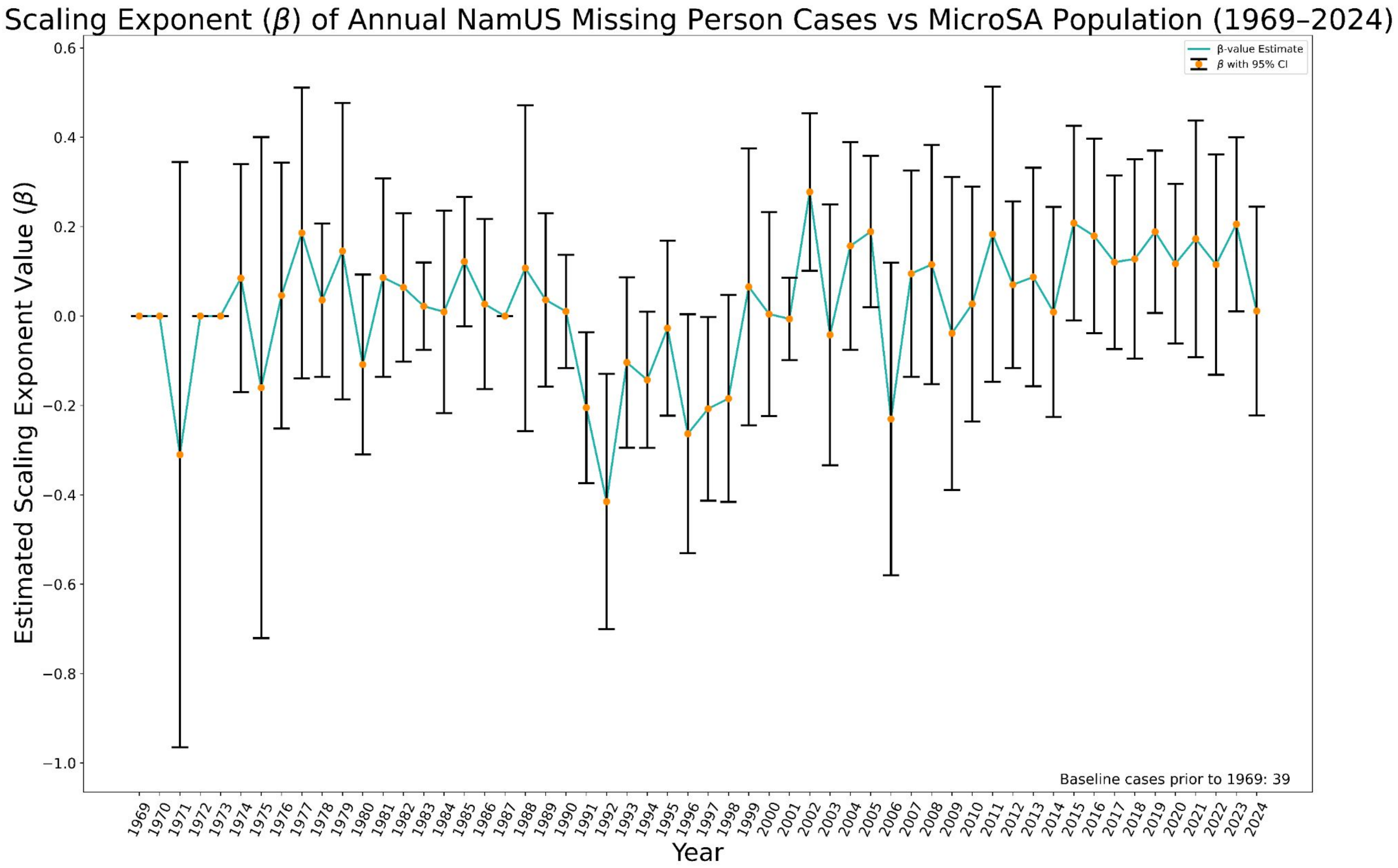
NYU


Worst Year (Annual): 1973 (R² = 0.075)
β = 0.109; CI: [-0.007, 0.226]
γ = -0.538; CI: [-1.224, 0.148]
Data Points
Regression Line
95% CI
log10(Cases)
log10(Population)
Best Year (Annual): 1991 (R² = 0.562)
β = 0.464; CI: [0.384, 0.543]
γ = -2.447; CI: [-2.909, -1.985]
Data Points
Regression Line
95% CI
log10(Cases)
log10(Population)

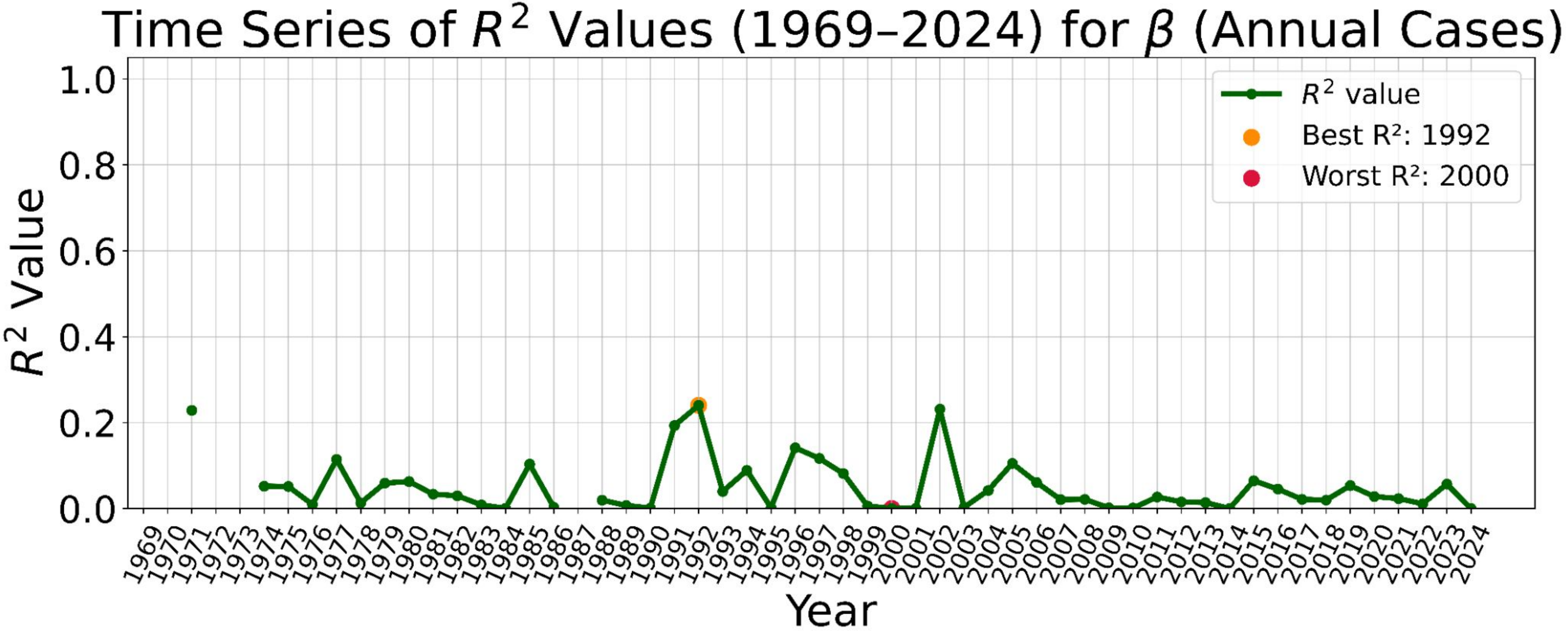
NYU

MicroSA-level

NYU

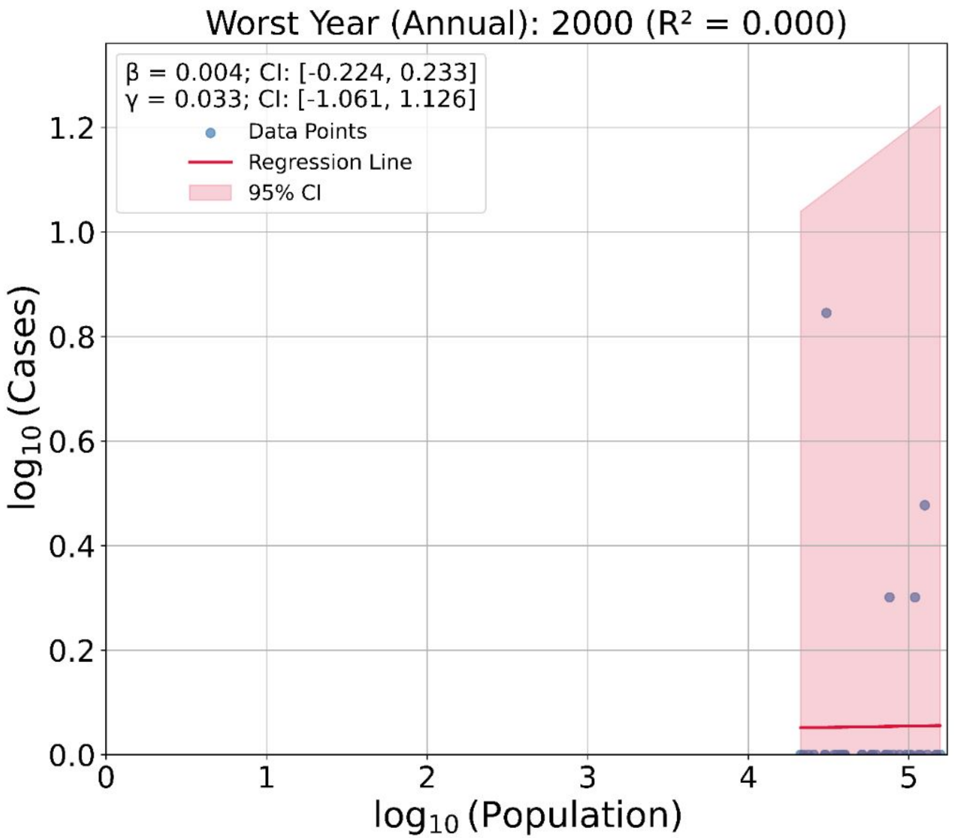


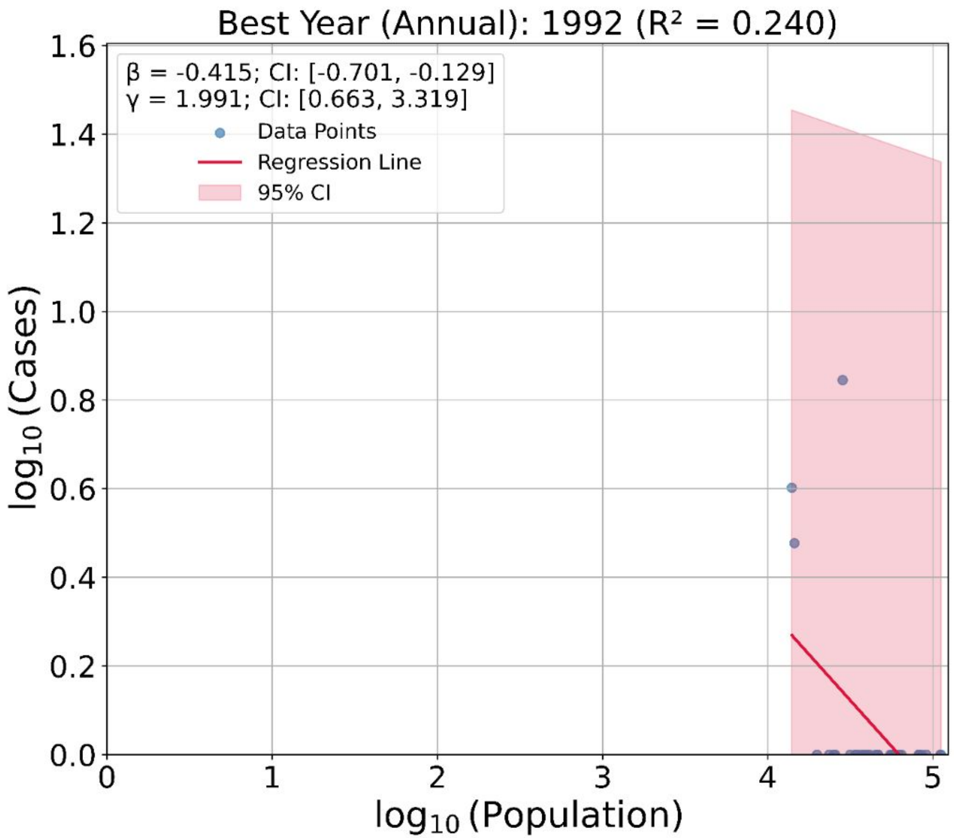
Scaling Exponent (β) of Annual NamUS Missing Person Cases vs MicroSA Population (1969–2024)
β-value Estimate
β with 95% CI
Estimated Scaling Exponent Value (β)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
Baseline cases prior to 1969: 39
1969 1970 1971 1972 1973 1974 1975 1976 1977 1978 1979 1980 1981 1982 1983 1984 1985 1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024
Year



Time Series of R^2 Values (1969–2024) for β (Annual Cases)
R^2 Value
1.0
0.8
0.6
0.4
0.2
0.0
R^2 value
Best R²: 1992
Worst R²: 2000
1969 1970 1971 1972 1973 1974 1975 1976 1977 1978 1979 1980 1981 1982 1983 1984 1985 1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024
Year

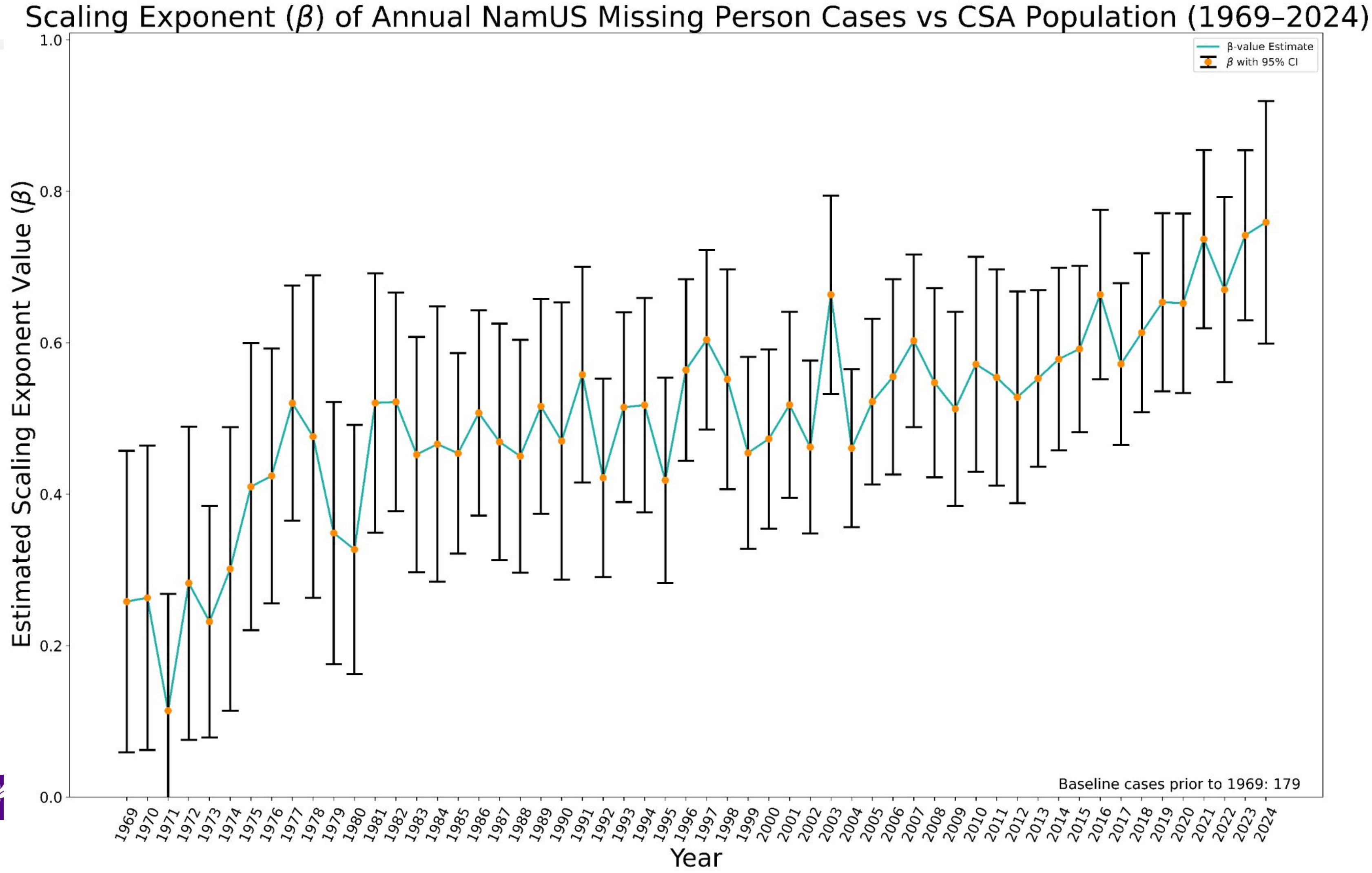
NYU

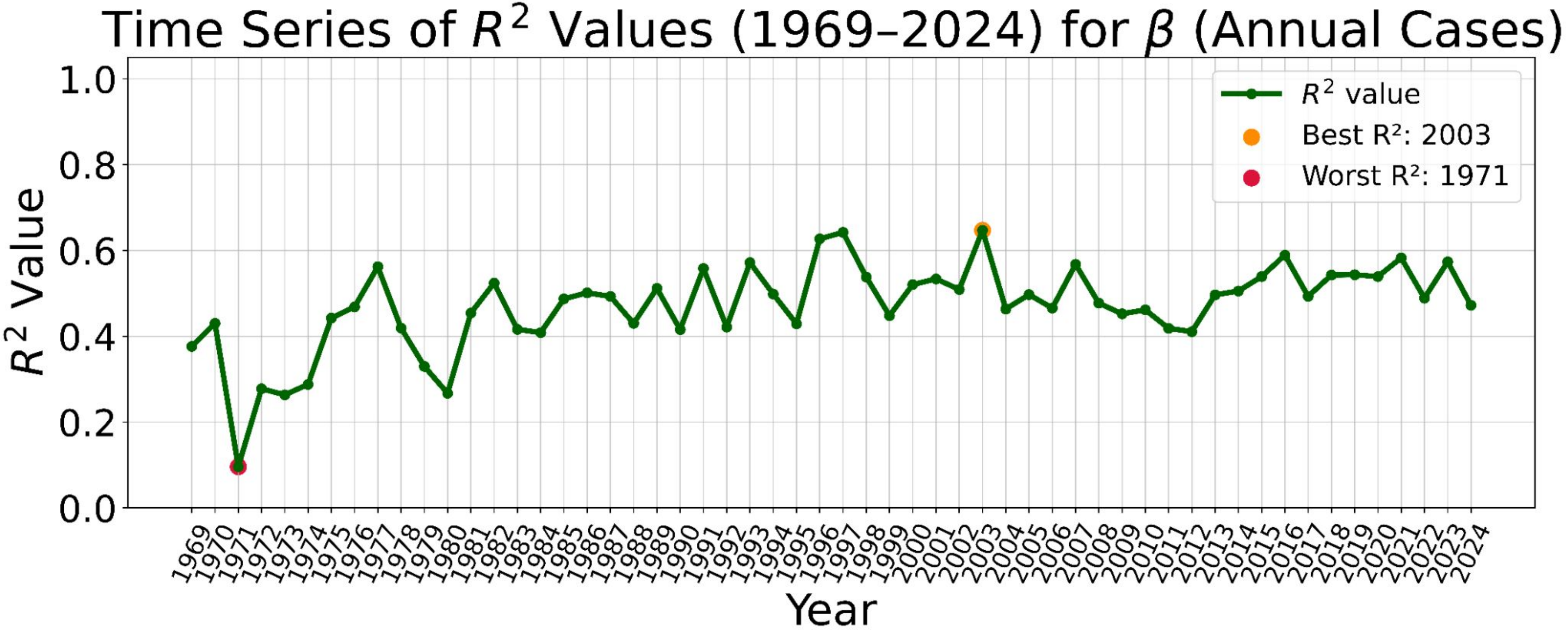

Worst Year (Annual): 2000 (R² = 0.000)
β = 0.004; CI: [-0.224, 0.233]
γ = 0.033; CI: [-1.061, 1.126]
Data Points
Regression Line
95% CI
log10 (Cases)
log10 (Population)


Best Year (Annual): 1992 (R² = 0.240)
β = -0.415; CI: [-0.701, -0.129]
γ = 1.991; CI: [0.663, 3.319]
Data Points
Regression Line
95% CI
log10 (Cases)
log10 (Population)

CSA-level

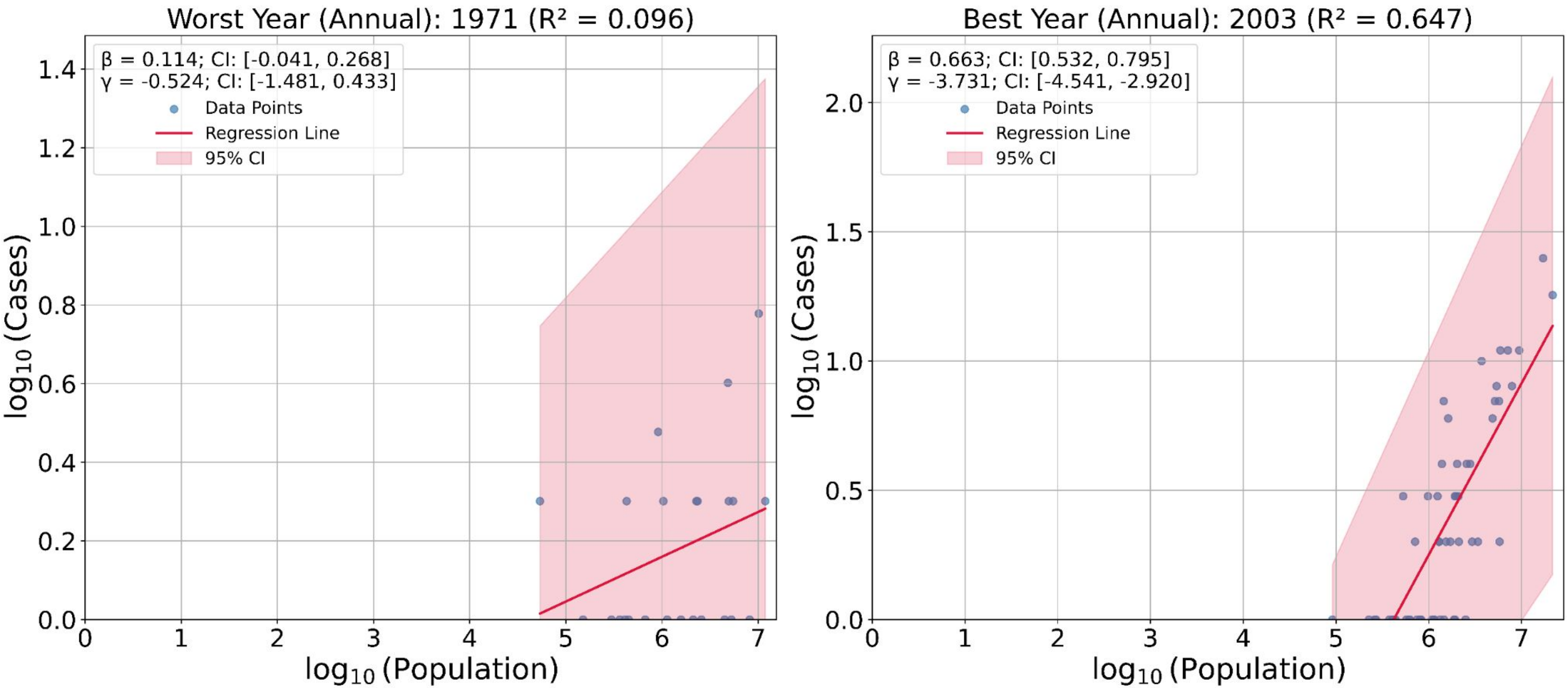
NYU


Scaling Exponent (β) of Annual NamUS Missing Person Cases vs CSA Population (1969–2024)
β-value Estimate
β with 95% CI
Estimated Scaling Exponent Value (β)
1.0
0.8
0.6
0.4
0.2
0.0
Baseline cases prior to 1969: 179
1969
1970
1971
1972
1973
1974
1975
1976
1977
1978
1979
1980
1981
1982
1983
1984
1985
1986
1987
1988
1989
1990
1991
1992
1993
1994
1995
1996
1997
1998
1999
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
Year



Time Series of R^2 Values (1969–2024) for β (Annual Cases)
R^2 value
Best R²: 2003
Worst R²: 1971
R^2 Value
Year
1.0
0.8
0.6
0.4
0.2
0.0
1969 1970 1971 1972 1973 1974 1975 1976 1977 1978 1979 1980 1981 1982 1983 1984 1985 1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024

NYU


Worst Year (Annual): 1971 (R² = 0.096)
β = 0.114; CI: [-0.041, 0.268]
γ = -0.524; CI: [-1.481, 0.433]
Data Points
Regression Line
95% CI
log10(Cases)
log10(Population)
Best Year (Annual): 2003 (R² = 0.647)
β = 0.663; CI: [0.532, 0.795]
γ = -3.731; CI: [-4.541, -2.920]
Data Points
Regression Line
95% CI
log10(Cases)
log10(Population)

NYU

Mexico INEGI Cases

NYU

Mexico Missing Persons Cases Data Characteristics

- We have **129,830** available records, with % **incomplete** by column:
    - Victim ID: **0.00%**
    - Origin Agency: **0.00%**
    - Date of Birth: **59.31%**
    - Sex: **37.05%**
    - Date of Incidence: **43.13%**
    - Date of Report: **41.52%**
    - Victim Status: **96.13%**
    - State: **2.26%**
    - State ID: **0.00%****
    - Municipality: **40.92%**
    - Municipality ID: **0.00%****

incomplete entries are 'MISSING', 'UNKNOWN', or 'CENSORED'*

GEOID IDs that are incomplete have a specific code

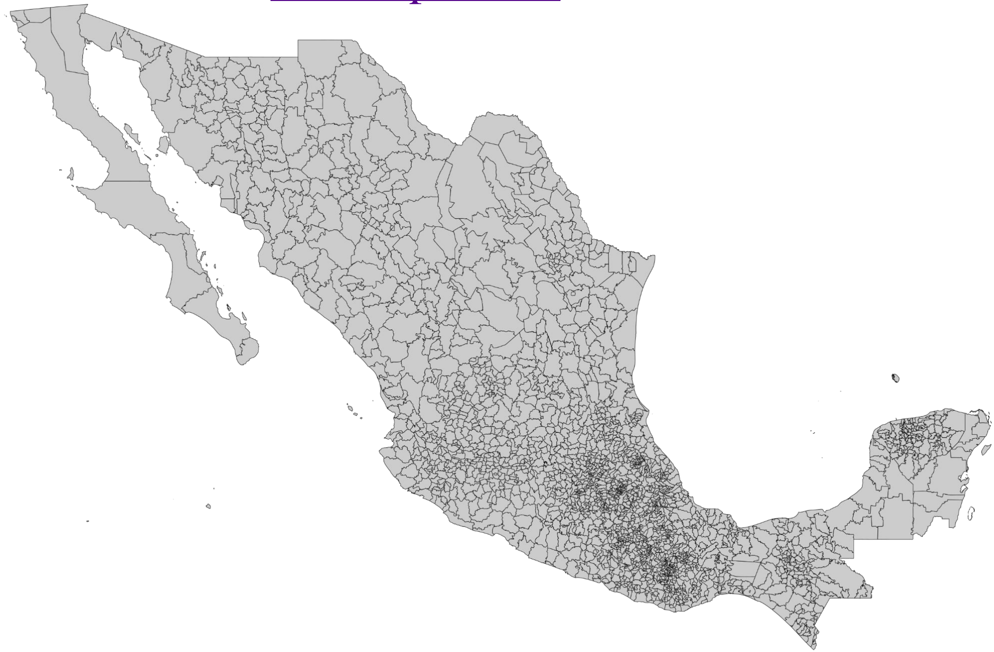
- Guide to Mexican Administrative Divisions
- The Mexico-American GEOID equivalencies by average population size **seem** to follow:
    - Mexican 'States' == American 'States'
    - Mexican 'Municipalities' == American 'Counties?' & American 'MSAs/CBSAs'
        - Extremely wide range of population values
        - Includes:
            - 'Ciudades'
            - 'Villas'
            - 'Pueblos'
            - …
    - Mexican 'Localities' == American 'Blocks?'
    - Mexico City 'Boroughs' == American 'Counties'

Mexico by GEOIDS

<u>States:</u>

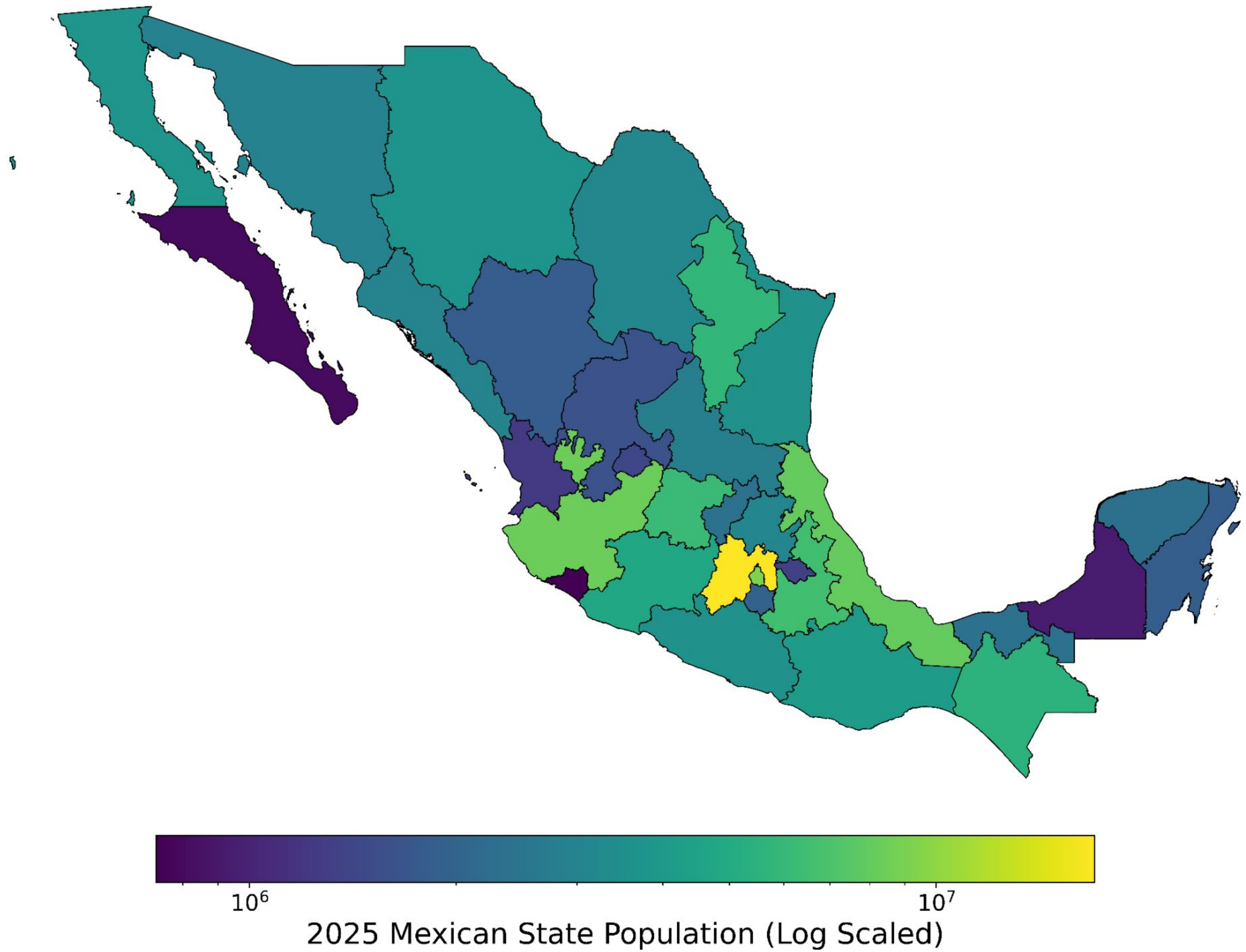
By Eddo - This W3C-unspecified vector image was created with Adobe Illustrator., CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=14581939

<u>Municipalities:</u>

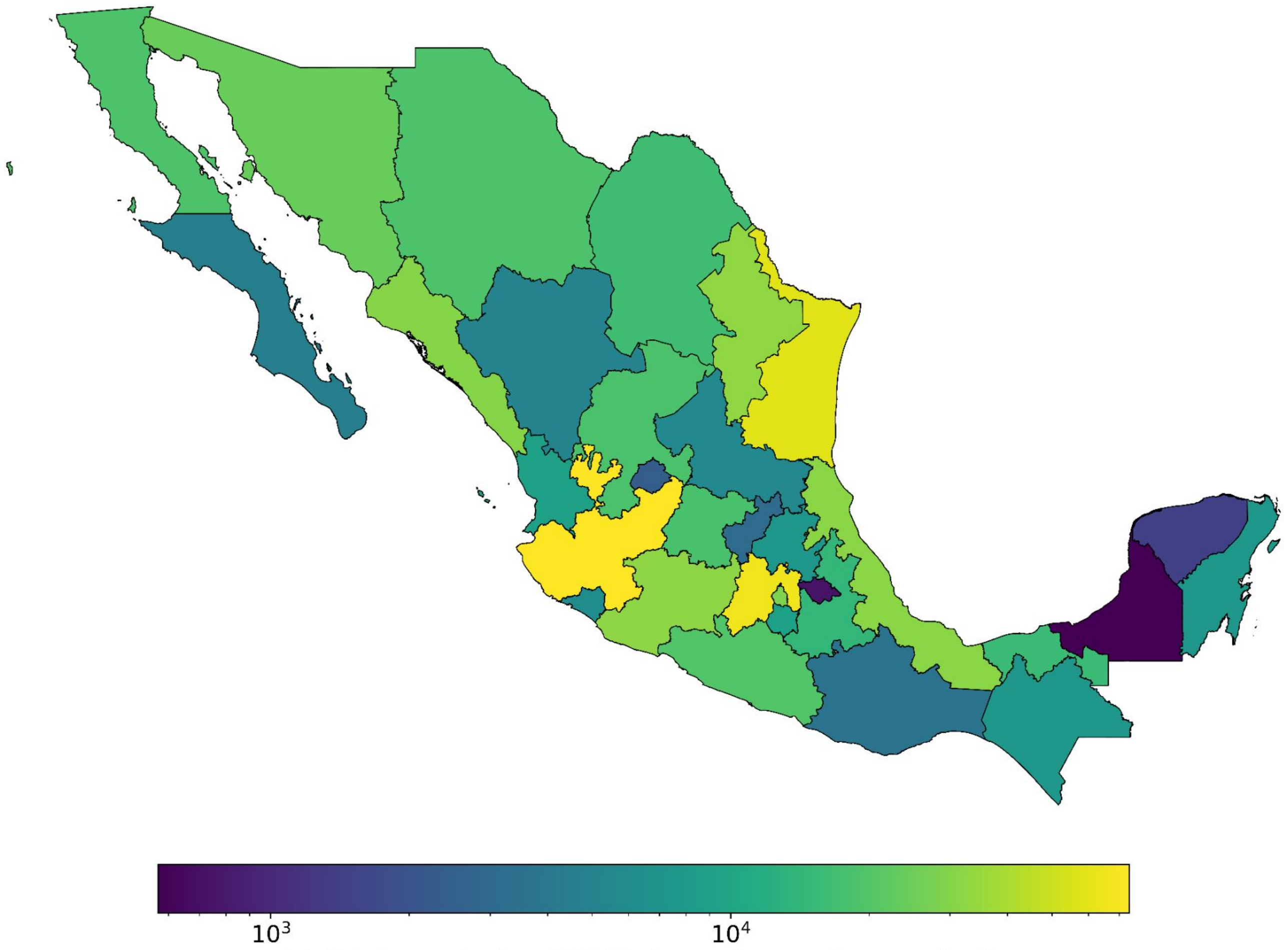


By Goran tek-en, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=96160260

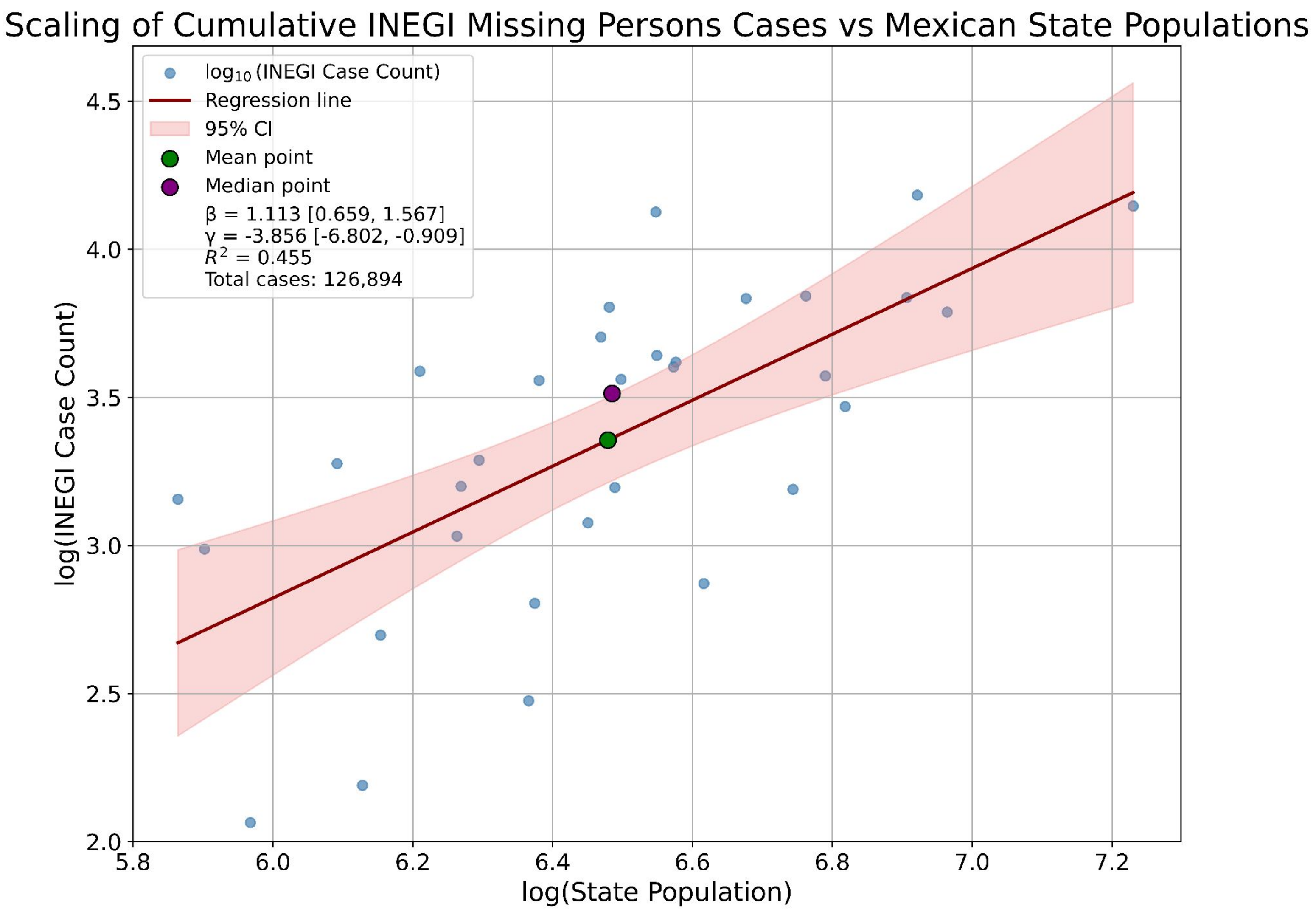
2025 Mexican State Population Choropleth (Log Scaled)



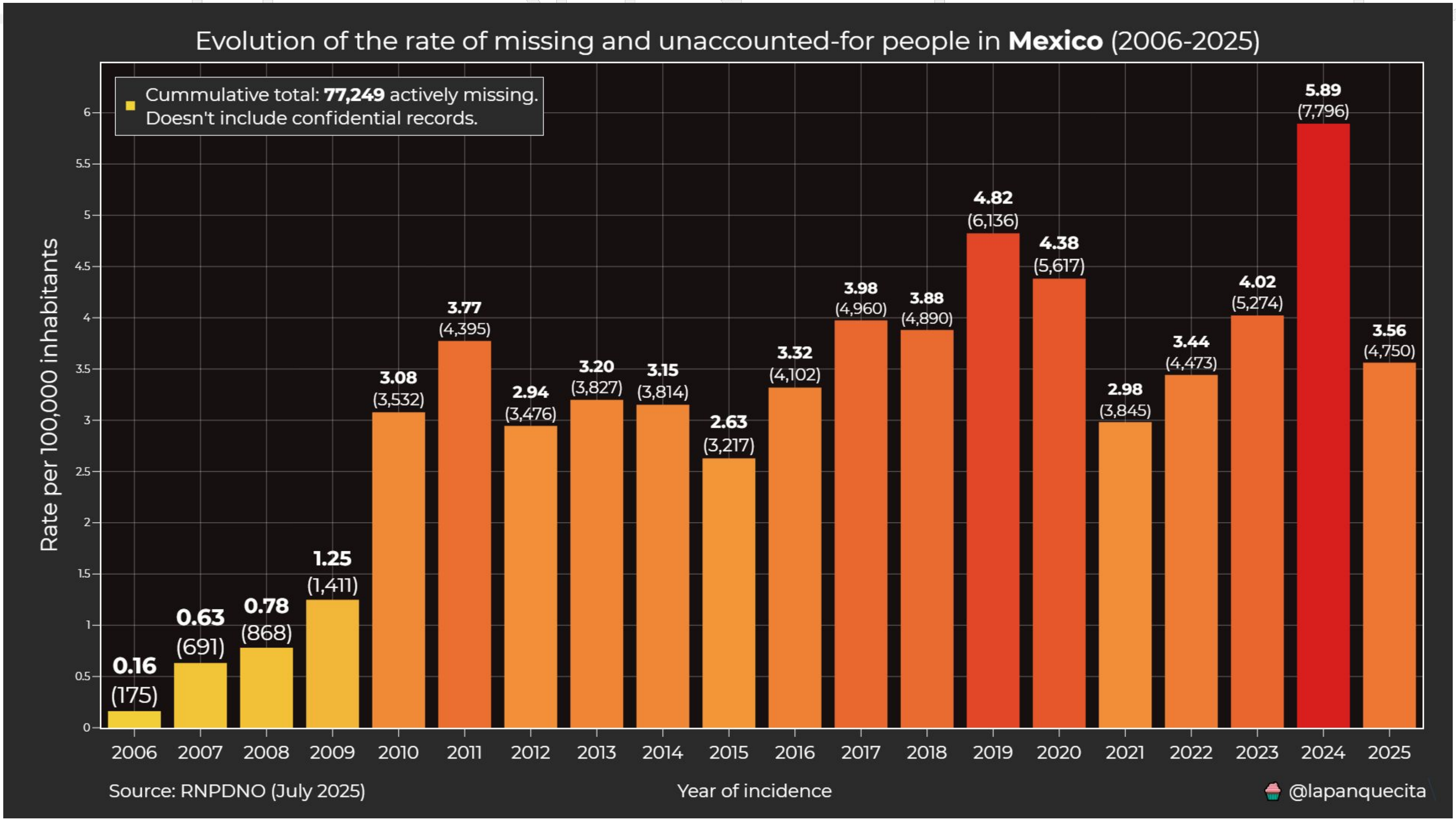
Valid INEGI Missing Persons Cases by Mexican State (Log Scaled)



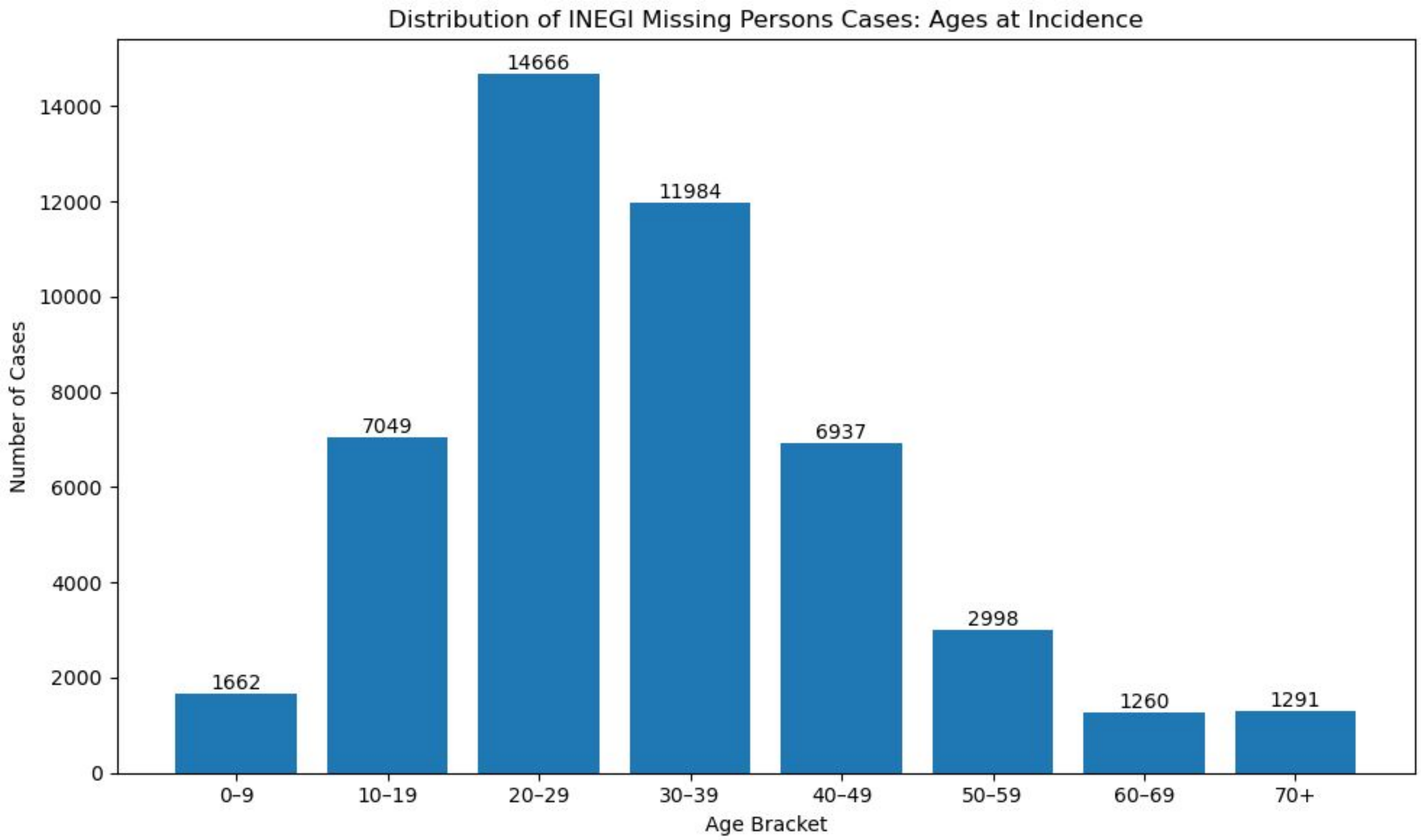
NYU

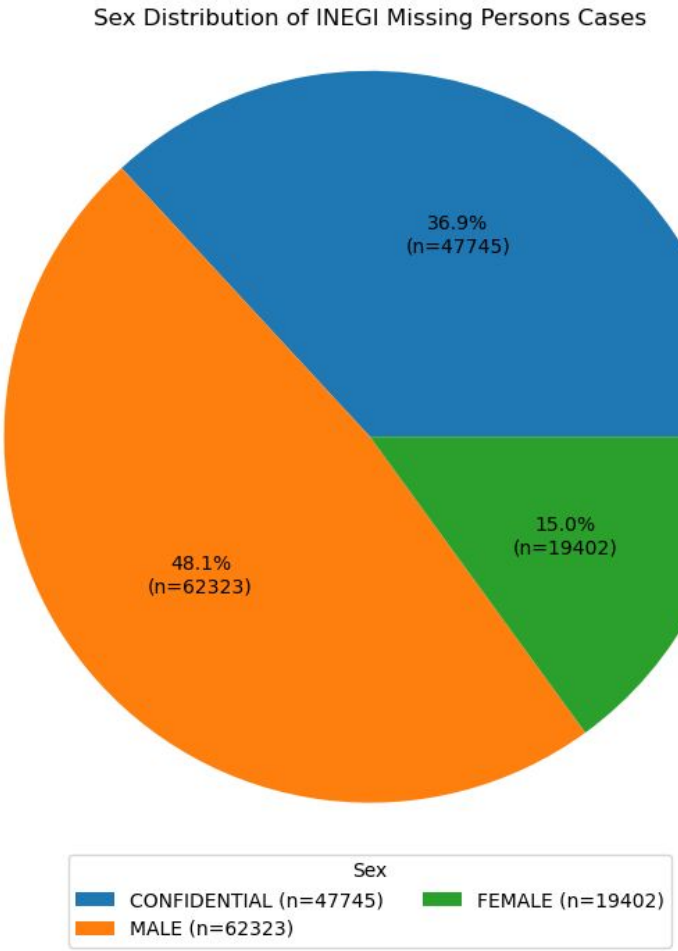


Scaling of Cumulative INEGI Missing Persons Cases vs Mexican State Populations
log10 (INEGI Case Count)
Regression line
95% CI
Mean point
Median point
β = 1.113 [0.659, 1.567]
γ = -3.856 [-6.802, -0.909]
R2 = 0.455
Total cases: 126,894
log(INEGI Case Count)
log(State Population)
2.0
2.5
3.0
3.5
4.0
4.5
5.8
6.0
6.2
6.4
6.6
6.8
7.0
7.2



Evolution of the rate of missing and unaccounted-for people in **Mexico** (2006-2025)
Cummulative total: **77,249** actively missing.
Doesn't include confidential records.
Rate per 100,000 inhabitants
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
5.5
6
0.16 (175)
0.63 (691)
0.78 (868)
1.25 (1,411)
3.08 (3,532)
3.77 (4,395)
2.94 (3,476)
3.20 (3,827)
3.15 (3,814)
2.63 (3,217)
3.32 (4,102)
3.98 (4,960)
3.88 (4,890)
4.82 (6,136)
4.38 (5,617)
2.98 (3,845)
3.44 (4,473)
4.02 (5,274)
5.89 (7,796)
3.56 (4,750)
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Source: RNPDNO (July 2025)
Year of incidence
@lapanquecita


Distribution of INEGI Missing Persons Cases: Ages at Incidence
Number of Cases
0
2000
4000
6000
8000
10000
12000
14000
1662
7049
14666
11984
6937
2998
1260
1291
0-9
10-19
20-29
30-39
40-49
50-59
60-69
70+
Age Bracket


Sex Distribution of INEGI Missing Persons Cases
36.9%
(n=47745)
48.1%
(n=62323)
15.0%
(n=19402)
Sex
CONFIDENTIAL (n=47745)
MALE (n=62323)
FEMALE (n=19402)

NYU